Investify

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Session Fall 2026 (FYP-1 Mid-Evaluation)

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11 October 2026

Students' Submission

We submit this report for the FYP-1 Mid-Evaluation of *Investify*, Fall 2026, Department of Software Engineering, FAST–NUCES Islamabad. It documents our implemented modules, evaluation results and remaining development plan. We accept responsibility for the submission and its stated contributions.

Ahmed Javaid (22I-2530) Signature: _____

Rafay Ul Haq (22I-2520) Signature: _____

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Date: _____

Anti-Plagiarism Statement

We declare that sources, reusable software, pretrained models and external data used in this project are acknowledged and referenced. The report distinguishes our implementation and integration work from external components. Development assistance is disclosed in the tool-use statement in Appendix G. We accept responsibility for accurate attribution and for the material submitted.

Ahmed Javaid (22I-2530) Signature: _____

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Date: _____

Authors' Declaration

We declare that this report describes the work delivered by the Investify project team for FYP-1 Mid-Evaluation. The member responsibilities and delivered modules are recorded in Chapter 9. Scientific results, limitations and external contributions are identified accurately. Each member accepts responsibility for understanding and defending their contribution and the integrated system.

Ahmed Javaid (22I-2530) Signature: _____

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Abstract

Investify is an educational equity decision-support prototype combining technical indicators, financial-news sentiment and explanations checked against model evidence. Its historical pipeline uses exchange-session alignment, causal features and purged chronological evaluation. A verifier checks displayed claims independently of structured payloads and supplies a checked fallback.

An NVDA experiment covers 1,509 price sessions from January 2018 to December 2023 and 6,768 news rows. Clean five-session macro-F1 is 0.2843 versus a random baseline of 0.3287; a deliberate future-looking feature triggers the leakage control. Independent isotonic calibration produces only Buy decisions. A multiclass temperature revision preserves the raw Buy/Hold/Sell distribution but exposes near-uniform probabilities, so low-confidence cases abstain. Original social classifiers are compared on 24 held-out reference-labelled posts. Risk profiling implements thirteen Grable–Lytton items with capacity constraints and a direction-preserving confidence policy. The live interface demonstrates evidence inspection, profile calculation, explanation rejection and recovery. These results establish implemented controls and traceability; useful prediction, student-model adaptation and participant acceptance remain later evaluation goals.

Acknowledgments

We acknowledge Mr. Irfan Ullah's role as project supervisor and the Department of Software Engineering for the FYP assessment guidance and report template. We also acknowledge the authors of the cited datasets, models and software libraries that support the implementation. Their contributions are identified in the report and references.

Executive Summary

Investify addresses two practical problems: historical evaluations may use information unavailable at decision time, and a fluent explanation may misstate the evidence behind a prediction. Our prototype combines a temporal evaluation harness with independently checked explanations. The interface exposes the complete technical/news feature context, validates visible text and structured claims separately, and preserves both a rejected candidate and its checked replacement.

The historical pipeline currently evaluates NVDA over 2018–2023. FNSPID was retained as the original comparison corpus, but its midnight-only timestamps and uneven ticker coverage limited precise session alignment. Alpaca/Benzinga provides the new timestamped news snapshot. The source change, selected price basis, data interval and remaining vendor-vintage limitation are documented. The seven-company product universe remains an integration target rather than a completed claim.

Prediction quality remains weak. The clean five-session walk-forward result is below its random comparator. A deliberately centred indicator produces positive lift, demonstrating that the harness detects a known leakage fault. The original isotonic mappings improve Brier score while forcing every decision to Buy. We diagnosed that mechanism and implemented a new global-temperature calibration experiment without replacing the original records. It preserves varied raw classes but approaches uniform confidence; a confidence gate therefore withholds unsupported directional suggestions. Calibration cannot manufacture predictive information.

The risk layer now uses the thirteen-item Grable–Lytton scoring key alongside five capacity factors. Capacity and tolerance reconcile restrictively. The resulting risk allowance controls the confidence required for a suggestion, not the relative sentiment/technical weight. Increasing allowance cannot turn Buy into Sell or Sell into Buy. The profile questionnaire, uncertainty response and explanation-failure path execute live; model outputs and research studies are recorded.

Ahmed Javaid delivered the temporal-data, leakage and CI/delivery work; Rafay Ul Haq delivered the interface, risk and sentiment work; Soban Ahmad delivered the engine, attribution and explanation-verification work. Responsibilities are equal for this evaluation, and members continue practising their modules and the integrated demonstration. The

report includes detailed use cases, requirements grouped by module, acceptance checks, individual test cases, revised design models and a dated engineering plan. Hosted Windows/Ubuntu checks and verified artifact delivery provide reproducibility evidence. Later work addresses broader data integration, improved prediction, language-model adaptation, production serving and participant evaluation.

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Abbreviations

FYP Final Year Project.

FR / NFR Functional / non-functional requirement.

CI/CD Continuous integration / continuous delivery.

SHAP Shapley additive explanations.

LLM Large language model.

QLoRA Quantized low-rank adaptation.

UTC Coordinated Universal Time.

SIP Securities Information Processor market-data feed.

GL Grable–Lytton tolerance scoring.

NLL Negative log likelihood.

Chapter 1

Introduction

1.1 Project Overview

Investify is an educational equity decision-support system. It combines technical indicators and financial-news sentiment, presents the evidence behind a model output, and checks explanations against that evidence. The prototype serves historical research cases; it does not connect to brokerage accounts or execute orders.

1.2 Problem Statement

An investor must interpret price indicators, news, uncertainty and personal risk constraints. A Buy/Hold/Sell label alone does not show which inputs influenced the result. Historical evaluations can also overstate performance when features or training outcomes become available after the decision time. Finally, fluent explanations can misstate numerical values or attribution directions. We address temporal evaluation and explanation consistency as explicit engineering problems.

1.3 Objectives and Contributions

Our implemented contributions are a leakage-audited evaluation pipeline and a record-grounded explanation-verification path. The former checks availability, chronological processing and deliberate faults; the latter exposes all available feature evidence, checks displayed and structured claims independently, and verifies recovery. Risk profiling and uncertainty gating connect these controls to the interface. News/social divergence remains a planned ablation rather than a completed result.

1.4 Scope and Modules

The full product targets the proposal's seven-company equity universe. The current research path is bounded to NVDA over 2018–2023, with a separate MSFT social-model comparison. The exposed prediction horizon is five trading sessions. Figure 5.1 distinguishes the six product modules and their current interfaces. Broader social integration, trained-student explanations and production serving remain later work.

Table 1.1: Product modules and current Mid-1 boundaries.

Module	Delivered behavior	Remaining integration
Investor profiling	Five capacity factors, thirteen tolerance items and restrictive reconciliation	Population-specific wording/suitability evaluation
Sentiment and data	Versioned prices/news, FinBERT features and separate social comparison	Multi-company coverage and news/social divergence
Predictive model	XGBoost records, calibration diagnosis and multiclass revision	Useful predictive skill and broader validation
Explainability	Complete feature context and native TreeSHAP	Participant interpretation of explanations
Explanation verifier	Independent claim checks, rejection and checked fallback	Larger real-generation comparison
Interface and service	Live Streamlit workflow and automated artifact delivery	Production service, follow-up chat and hosting

1.5 Stakeholders and Success Criteria

The retail learner inspects evidence and uncertainty; the operator maintains data and runtime health; the supervisor assesses the implementation and the team defends its modules. Success is assessed through causal-alignment checks, deliberate-fault detection, held-out prediction/calibration results, explanation consistency, reproducible setup and later participant tasks. Software correctness and predictive usefulness are reported separately.

1.6 Constraints, Privacy and Accessibility

The historical demonstration uses versioned public market material and local profile answers. API credentials remain outside source control and report artifacts. Profile data is kept in the local session and is not sent to a brokerage. Text labels accompany colors, controls are keyboard accessible and evidence remains inspectable without a GPU. The interface withholds directional suggestions when predictive confidence is insufficient. Population-specific risk assumptions, data vintage and eventual user acceptance require further evaluation.

Chapter 2

Research and Technical Background

2.1 Financial Text and Prediction

FinBERT adapts language representations to financial sentiment and provides positive, neutral and negative scores [1, 2]. Investify derives news polarity from those scores and combines it with technical indicators. The language model is an input component, not a stock-return oracle. FNSPID supplies historical financial text [3]; source coverage and timestamp precision must be audited before aligning it to tradable sessions.

2.2 Model Selection by Text Register

Financial headlines and informal social posts differ in language and context. FinBERT remains the news model. CryptoBERT was proposed for social content, but its crypto-domain training does not establish equity-tweet transfer [4]. Cardiff's Twitter sentiment model provides a relevant alternative with explicit mention/URL preprocessing [5]. Both original checkpoints were evaluated on the same frozen references. Cardiff's exploratory lead and the added sanity probes are reported in Chapter 8; changing the social candidate is documented as an evaluated alternative to the proposal, not a replacement for FinBERT.

2.3 Attribution and Explanation Checking

TreeSHAP decomposes a tree model's raw score into a base value and feature contributions [6]. These values are not calibrated probabilities or economic causes. Explanation research distinguishes faithfulness from self-consistency [7]; Investify therefore makes a narrow, testable claim: supported statements must match the selected record. FActScore motivates atomic-claim accounting [8], while RARR illustrates evidence-based correction

[9]. Our contribution is a bounded implementation with visible-text/JSON independence, coverage gates, retained rejection and independently checked fallback.

Table 2.1: Related methods and their role in Investify.

Method	Relevant capability	Project distinction
Financial sentiment models	Text-domain classification	Explicit news/social routing and transfer study
TreeSHAP	Raw-score attribution	Complete feature context and additivity checks
Atomic factuality evaluation	Supported-claim accounting	Numerical record, required visible facts and strict vocabulary
Evidence-based revision	Correct unsupported text	Reject, retain failure, then verify deterministic recovery
Chronological financial evaluation	Future-independent estimation	Endpoint purge, explicit gap and permanent fault canaries

2.4 Multiclass Calibration

Independent one-versus-rest isotonic maps need not preserve the multiclass ranking. The fresh maps make Buy's minimum exceed the other classes' maxima, creating constant output. Temperature scaling instead divides all class margins by one positive parameter and normalizes jointly [10]. We fit that parameter on earlier calibration outcomes only. The observed fit approaches uniform confidence, so the interface abstains rather than treating a preserved raw class as a usable recommendation. This is a post-analysis correction. Parameters use calibration outcomes only; evaluation cases were already examined, so the new score is diagnostic rather than a first-unseen performance claim.

2.5 Risk Capacity and Tolerance

Capacity describes the ability to bear loss; tolerance describes willingness. Grable–Lytton uses thirteen items with a published scoring key and range 13–47 [11]. We retain the scoring and display concise paraphrases, alongside five capacity factors and explicit protective caps. The more restrictive capacity/tolerance tier binds. A normalized allowance controls required confidence; it is no longer a sentiment weight. Local questionnaire wording and the project confidence policy still require population-specific evaluation.

2.6 Market Mechanisms and Data Vintage

Stocks represent ownership interests and may respond to company results, financing conditions, expectations and market-wide events [12]. Technical indicators describe recent price/volume structure; sentiment describes the text, not its truth. SHAP describes how a fitted model used an input, not why the market moved.

NVIDIA's four-for-one split began split-adjusted trading on 20 July 2021 [13]; the ten-for-one split began on 10 June 2024 [14]. The second event occurs after our 2018–2023 window, but a 2026 vendor snapshot can retroactively rescale earlier prices and volumes. A split changes share count and per-share scale, not holding value by itself. We state the consumed SIP split-adjusted basis and its acquisition vintage rather than describing it as original as-traded data. Dividends, action metadata and provider revisions remain distinct audit questions.

Exchange holidays and early closes require a declared timing policy [15]. The inherited baseline uses a fixed 16:00 New York cutoff, including half-days; it is not claimed as actual-close execution. Forward labels mature after five observed sessions. Label-end purging and the additional configured calendar-day gap are separate controls, explained in Chapter 6.

2.7 Teacher and Student Choice

The initial open-weight preference reflected anticipated restrictions on broad model distillation. For the current scoped construction pilot, a frontier assistant supplies explanatory examples; no competing general-purpose model or student adaptation is trained. The service restriction concerns competing models rather than a blanket prohibition on every educational example [16]. Phi-3-mini remains the planned original student and Mistral a fallback. The larger teacher/student comparison remains a later milestone. The construction method and development assistance are disclosed once in Appendix G.

2.8 Research Gaps

Useful prediction, unrestricted explanation quality, news/social divergence, multi-company generalization, population-specific risk interpretation and participant benefit remain open. The report separates those research questions from verified implementation behavior. Negative findings preserve their value as diagnostic evidence and guide the dated development plan.

Chapter 3

Proposal Feedback and Iteration Plan

3.1 Proposal Feedback

The project was approved with minor modifications. The panel requested clearer contributions and model choices, an explanation of the LLM's user benefit, technical/sentiment evidence, explicit metrics, an improved PoC, transparency and deeper market understanding. Appendix A links each action to the current response and remaining evaluation.

3.2 Iteration Strategy

Each increment delivers runnable behavior, source-linked tests and retained evidence. New experiments use new identities so earlier inputs and findings remain reproducible. The prototype distinguishes implemented controls from unproven predictive or participant outcomes.

3.3 Original Milestones and Current Progress

Table 3.1: Progress against the four proposal iterations.

Stage	Commitment	Current position
Mid-1, week 7	Data/calendar/features, social selection, baseline, leakage suite and thin interface	Core paths execute; report response and uncertainty/risk corrections included

Stage	Commitment	Current position
Final-1, week 12	Engine/calibration/SHAP, divergence, service and initial deployment	Engine/attribution built early; social integration and service work remain
Mid-2, week 18	Prompted explainer, filtered data, NFRs, parity, degraded operation and acceptance preparation	Verifier/adapter groundwork present; full measurements and integration remain
Final-2, week 24	QLoRA comparison, broader evaluation, participant sessions and external installation	Later development and evaluation

The cohort rubric places Mid-1 around week eight; proposal week numbers are relative planning milestones. One completed checkpoint does not imply a fixed whole-project percentage.

3.4 Work Breakdown and Delivered Responsibilities

The proposal allocation is retained, with each member delivering the relevant Mid-1 part into the integrated package. Chapter 9 distinguishes delivered work from later responsibilities. Members continue learning and practising the code they will defend. The shared presentation allocation remains one-third each.

Table 3.2: Proposal work division and Mid-1 delivery scope.

Member	Proposal responsibilities	Mid-1 share
Ahmed Javaid	Point-in-time harness, leakage tests, dual timestamps, parity and deployment	One-third; temporal/CI part delivered
Rafay Ul Haq	Risk profiling, sentiment, degraded operation, acceptance and frontend	One-third; interface/profile/study part delivered
Soban Ahmad	Engine, calibration, SHAP, verifier and QLoRA	One-third; model/checking part delivered

Deployment, participant acceptance and QLoRA are later responsibilities; the delivered column identifies the current evaluation scope.

3.5 Deviations and Recovery

FNSPID's timestamp precision and ticker gaps prompted a bounded NVDA study and a new Alpaca/Benzinga snapshot. FinBERT remains the headline model; Cardiff is an evaluated social alternative. The isotonic collapse prompted a separately identified multiclass revision and abstention guard. The original sentiment-weight profile was replaced in the main interface by complete tolerance scoring and a direction-preserving confidence policy. Earlier records and labels remain unchanged.

3.6 Dated Engineering Targets

Dates below are internal delivery targets; they do not replace university assessment appointments.

Table 3.3: Engineering milestones, ownership and target dates.

Target	Deliverable/check	Owner
14 Oct 2026	Mid-1 report response, risk/calibration corrections and revised demonstration	All three
21 Oct 2026	Broader calibration/baseline analysis and risk wording review	Soban, Rafay
4 Nov 2026	Multi-company data audit and news/social divergence integration	Ahmed, Rafay
18 Nov 2026	Service parity, container setup and degraded-operation tests	Ahmed, Soban, Rafay
2 Dec 2026	Integrated FYP-1 prototype and measured comparison evidence	All three
17 Feb 2027	Student checkpoint, prompted baseline and verified training-data expansion	Soban
17 Mar 2027	QLoRA and three-way explanation comparison	Soban, Rafay
14 Apr 2027	Participant tasks, external installation and final handover	All three

The next action for weak prediction is baseline/data/model analysis, not held-out tuning to force a positive result. If useful prediction remains absent, the temporal harness and explanation guard remain the evaluated contributions.

Chapter 4

Software Requirements

4.1 User Interactions

The learner selects a historical case, inspects evidence, completes a risk profile and checks an explanation. The operator installs the viewer and inspects input failures. Research views expose source-bound studies. The current workflow is historical and educational; the final conversational service remains later scope.

4.2 Use Case Diagram

Figure 4.1 uses actors outside the system boundary, oval use cases and labelled include/extend relationships. Study preparation is an operator task, not an unimplemented student actor.

4.3 Detailed Use Cases

4.3.1 UC-01: Select a Historical Decision

Actor Learner.

Precondition The viewer runs and a registered experiment exists.

Trigger Select an experiment and session.

Main flow The loader reads the declared artifact, validates schema and unique sessions, binds its digest and displays class/probabilities for the chosen case.

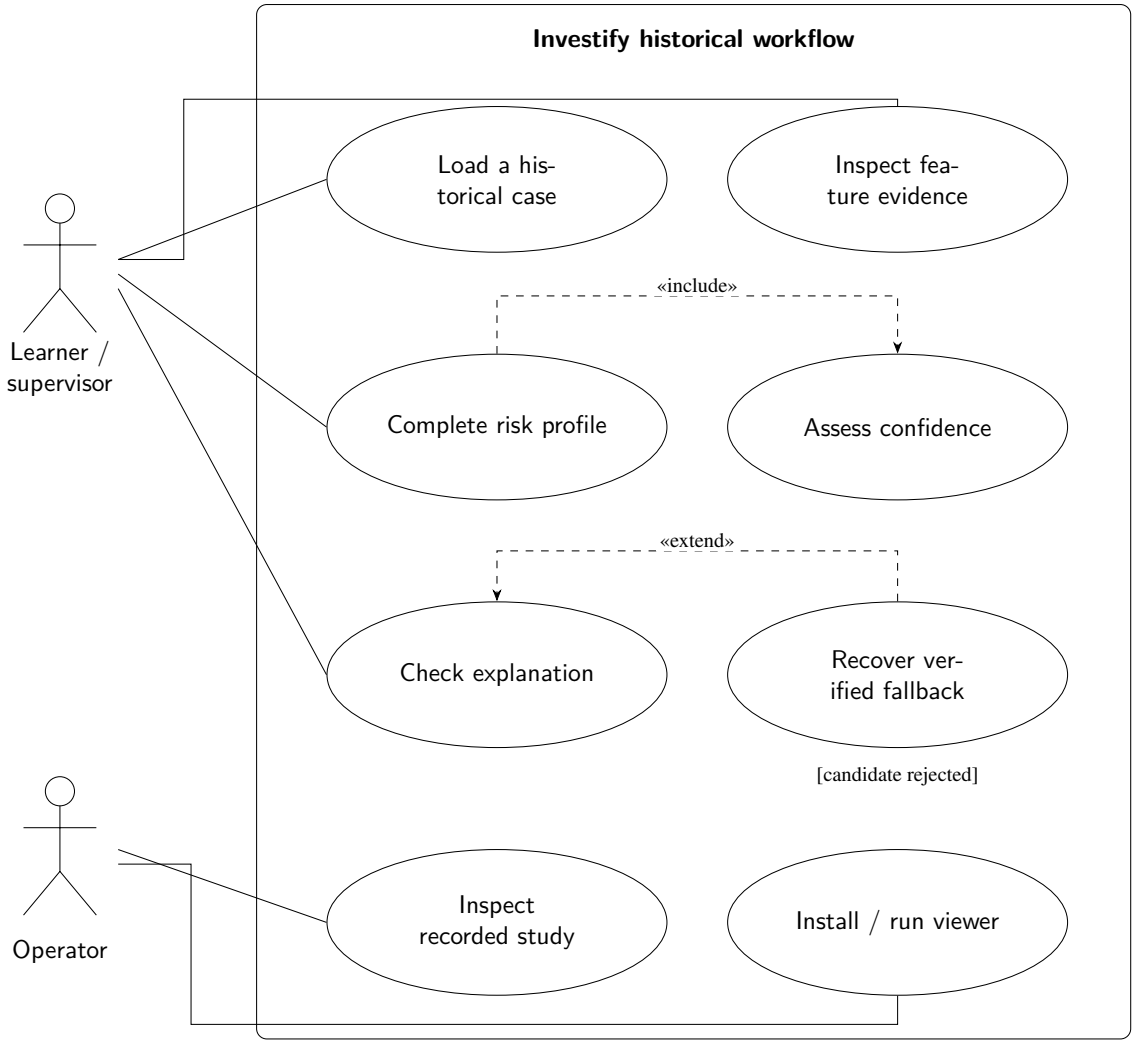


Figure 4.1: Implemented learner/operator use cases.

Exceptions Missing, malformed or changed evidence produces an error without a substituted decision.

Postcondition One source/date/digest identifies the active case; an obsolete explanation result is cleared.

4.3.2 UC-02: Inspect Technical and News Evidence

Actor Learner.

Precondition A valid case is selected.

Trigger Open Evidence and choose a family/feature.

Main flow Display the recorded value, maintained meaning, raw-score contribution and selected-class direction; expose all eleven inputs in detail.

Exceptions A missing value remains explicitly missing; an absent social signal is not fabricated.

Postcondition The learner can trace each displayed feature to the active record.

4.3.3 UC-03: Complete the Risk Profile

Actor Learner.

Precondition Profile form is available.

Trigger Submit five capacity and thirteen tolerance answers.

Main flow Validate all choices, sum published tolerance weights, calculate capacity/caps and reconcile the more restrictive tier. Display the binding axis and allowance R.

Exceptions Any unanswered/invalid choice withholds the result. Explicit fictional examples are labelled; Clear removes the profile.

Postcondition A versioned profile is held only in the local session.

4.3.4 UC-04: Apply the Confidence Policy

Actor Learner.

Precondition Valid case and complete profile.

Trigger Apply/change the profile or selected case.

Main flow Calculate $\tau(R) = 0.80 - 0.30R$ and compare the recorded class confidence. Withhold a directional suggestion below the threshold; never reverse its market direction.

Exceptions Low allowance/confidence produces Observe; an existing Hold signal remains non-directional.

Postcondition The profile outcome is displayed separately from unchanged model probabilities and SHAP.

4.3.5 UC-05: Check an Explanation and Recover

Actor Learner.

Precondition Valid selected record and current candidate text.

Trigger Click Check explanation.

Main flow Check visible and structured claims independently. If accepted, show the current candidate and report. If rejected, retain its failed report, render a deterministic fallback and check it independently.

Exceptions Contradictory, missing, malformed or unsupported claims fail. A fallback/checker failure displays an error, not a successful output.

Postcondition Only a result matching the current case/text is shown; edits invalidate it.

4.3.6 UC-06: Inspect a Recorded Study

Actor Learner or supervisor.

Precondition Retained study receipts and outputs are available.

Trigger Select leakage, sentiment or teacher study.

Main flow Validate identities and coverage, recompute supported summary values and expose sample/scope limits.

Exceptions Changed, incomplete or unavailable evidence withholds replacement scores.

Postcondition Recorded findings are displayed without a page-load model call.

4.3.7 UC-07: Install and Run the Demo

Actor Operator or group member.

Precondition Git/Python 3.11 and initial package access.

Trigger Follow the setup manual.

Main flow Clone source, create an environment, install pinned viewer dependencies, restore reviewed checkout bytes and start Streamlit on loopback.

Exceptions Missing runtime or occupied port produces an actionable setup error; existing processes are not stopped automatically.

Postcondition The historical workflow runs without model downloads or brokerage credentials.

4.4 Functional Requirements by Module

The IDs preserve traceability while grouping the implementation by responsibility. Profile/calibration revisions are explicitly versioned; the earlier blend remains a historical comparator.

4.4.1 Data, Decision and Evidence

Table 4.1: Data/decision functional requirements.

ID	Requirement	Acceptance
MID-FR-01	Load saved cases; refuse invalid/unavailable records	No synthetic replacement class
MID-FR-02	Expose all technical/news values, meanings and attribution	Five technical and six news inputs inspectable
MID-FR-07	Refuse scientific reruns when structural input/runtime preflight fails	Failure state precedes execution
MID-FR-19	Apply one positive multiclass temperature using earlier outcomes only	Raw argmax preserved; low confidence disclosed

4.4.2 Risk Profiling

Table 4.2: Profile and policy functional requirements.

ID	Requirement	Acceptance
MID-FR-12	Require complete capacity/tolerance answers and expose restrictive reconciliation	No defaults; binding caps/tier displayed
MID-FR-13	Keep historical comparator scaling chronological	Earlier fit only; unsuitable historical sessions refused
MID-FR-14	Keep current risk policy independent of model direction/probabilities	Allowance changes confidence requirement, not Buy/Sell direction
MID-FR-20	Implement all thirteen published tolerance scoring items	Explicit choices; total 13–47; min/max tests pass

4.4.3 Explanation and Recovery

Table 4.3: Explanation-verification functional requirements.

ID	Requirement	Acceptance
MID-FR-03/04	Check prose/JSON independently and reject contradictions/omissions	Correct payload cannot hide wrong prose
MID-FR-05/06	Retain failed report; reject unsupported strict-mode sentences	Candidate/fallback verdicts remain separate
MID-FR-08	Bound explicitly configured generator retries/timeouts	Failure is recorded; fallback labelled
MID-FR-09/11	Check maintained meanings, all-feature values/effects and raw-score scope	Both representations cover required facts

MID-FR-10	Preserve frozen selection and failed/raw attempts	No quality score without actual outputs

4.4.4 Research Studies

Table 4.4: Study provenance and comparison requirements.

ID	Requirement	Acceptance
MID-FR-15	Validate post/reference provenance, exclusions and chronological plan	Declared reference method and fixed identities
MID-FR-16/17	Require complete matching original-classifier outputs and mappings	Partial failures withhold paired metrics
MID-FR-18	Preserve actual teacher prose, independent checking and grouped exports	Test dates excluded; student training separate

4.5 Non-Functional Requirements

Table 4.5: Quantitative product targets and current measurement boundary.

ID	Target	Status
NFR-1	Decision/attribution/verification below 800 ms p95 on target VM	Target measurement remains
NFR-2/3	Sentiment at least 150 documents/s on specified GPU; CPU SHAP below 50 ms	Target environment measurement remains
NFR-4	Inference memory below 4 GB	Target measurement remains
NFR-5	Later QLoRA within four GPU hours on 16 GB envelope	Later training milestone
NFR-6	Explanation including verification below five seconds	Whole-request measurement remains

NFR-7/8	Nightly ingest below 30 minutes; walk-forward/search below 90 minutes	Operational measurement remains

The configured retry/socket bounds do not establish a whole-request latency guarantee. Reliability acceptance requires explicit unavailable/error states and current-identity results. Local routes reject malformed/nonfinite input and retain no brokerage credentials. Installation tests and hosted artifact delivery establish reproducibility of their scoped paths, not production security or participant comprehension.

4.6 Acceptance and Traceability

Positive cases and deliberate failures are paired: a correct explanation passes; its corrupted visible probability fails despite correct JSON; fallback passes independently; an edit clears the old verdict. The risk layer refuses incomplete inputs and preserves market direction. The source/test matrix in Appendix B maps the twenty requirement IDs to implementation and named checks.

Chapter 5

Software Design Specification

5.1 Product Architecture and Current Slice

Figure 5.1 shows the six-module product design and marks later integrations. The current slice executes record loading, evidence presentation, profile/policy calculation and explanation recovery. Recorded scientific outputs enter through explicit artifact contracts; model training is not a page-load action.

5.2 Components and Layers

The interface calls context, profile and verification services. Scientific routines produce the versioned records read by those services. Local adapters and study readers use declared identities rather than silent substitution.

5.3 Domain and Class Model

The class model includes attributes and operations used by the code. Multiplicities distinguish a collection of records, a record's contributions and candidate/verification relationships. Profile evaluation is a dependency on a record, not a mutation of classifier probabilities.

5.4 Data Flow and File Schema

The schema identifies experiment and composite case identities, keyed feature maps, candidate digests and result ownership. These are JSON/file/session contracts, not a claimed

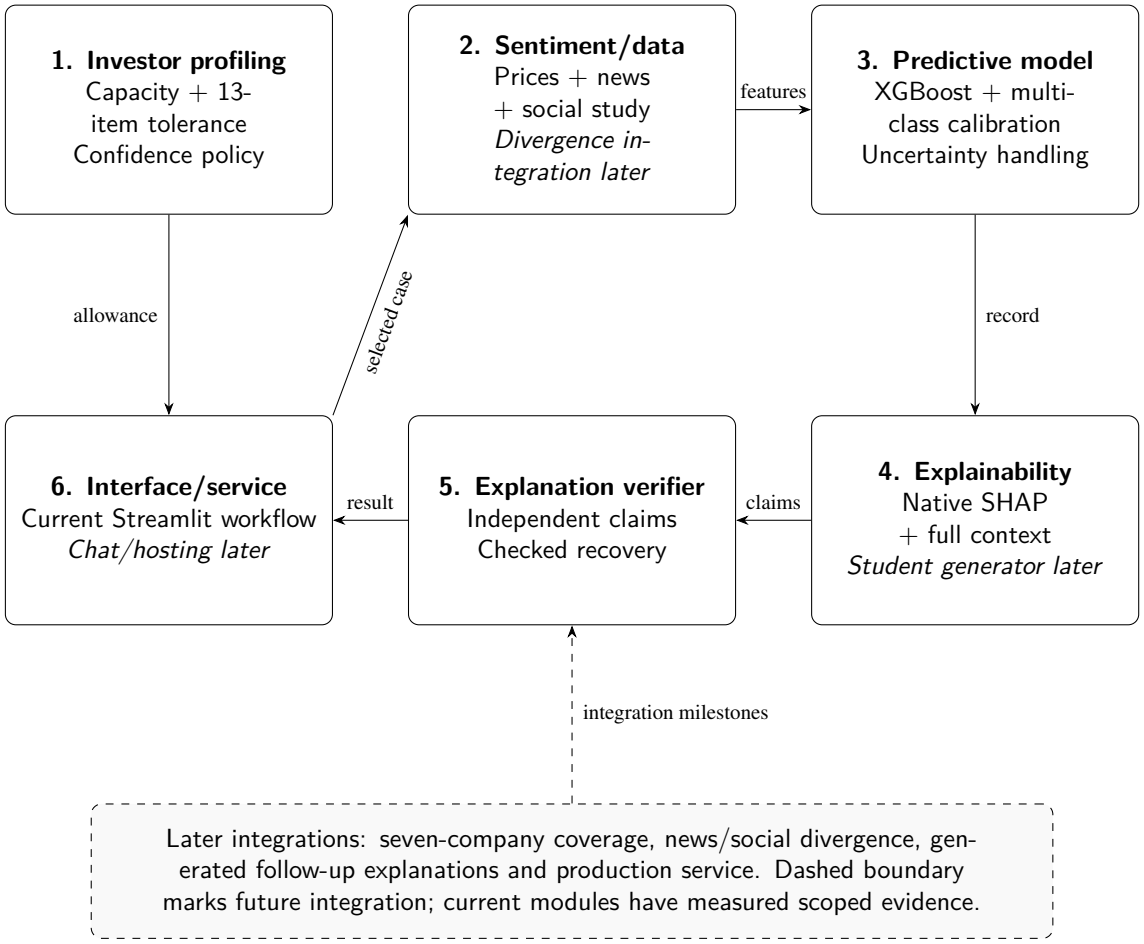
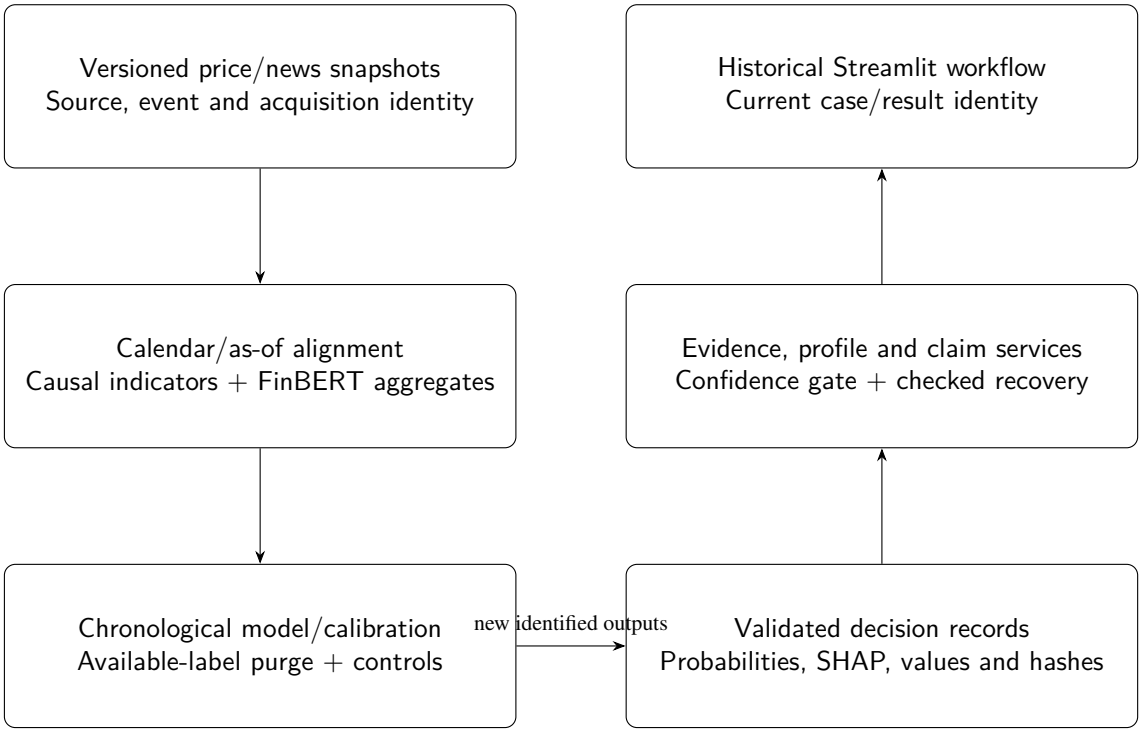


Figure 5.1: Six-module product architecture; dashed elements represent later integration.



Offline scientific execution produces records.
Interactive checking/profile logic uses them without retraining.

Figure 5.2: Current data-to-record and record-to-interface pipeline.

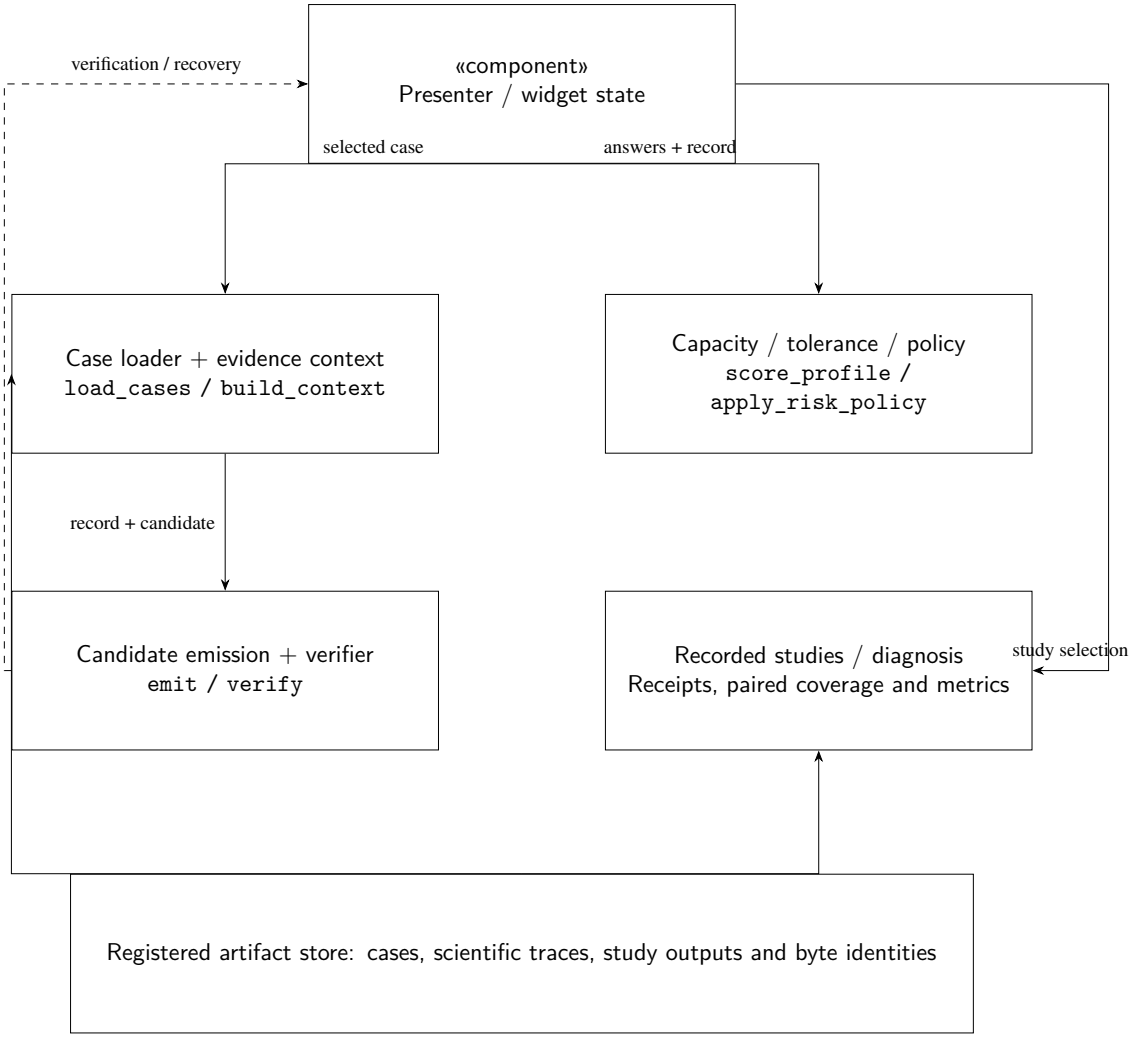


Figure 5.3: Implemented components and their interfaces.

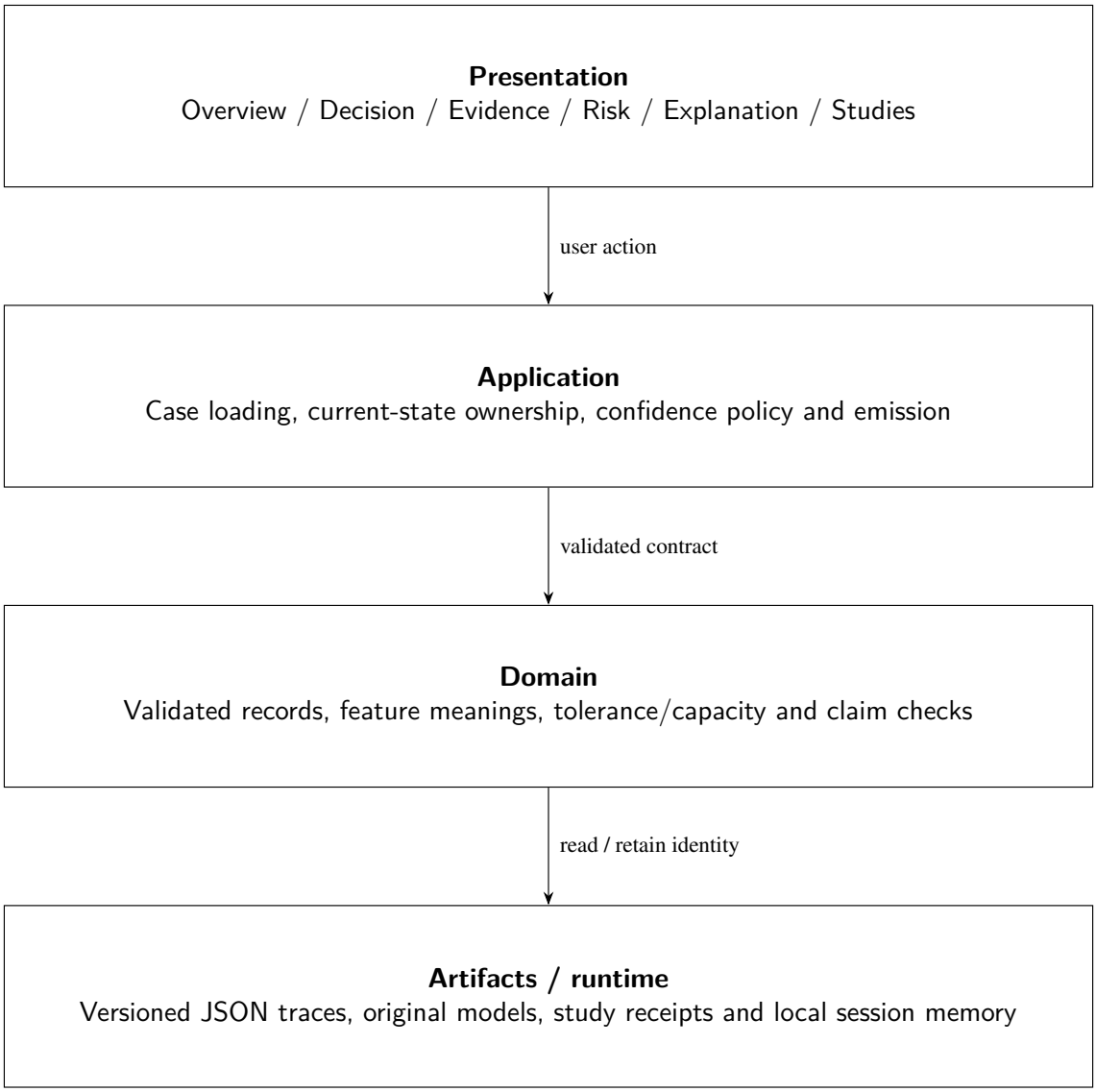


Figure 5.4: Presentation, application, domain and artifact layers.

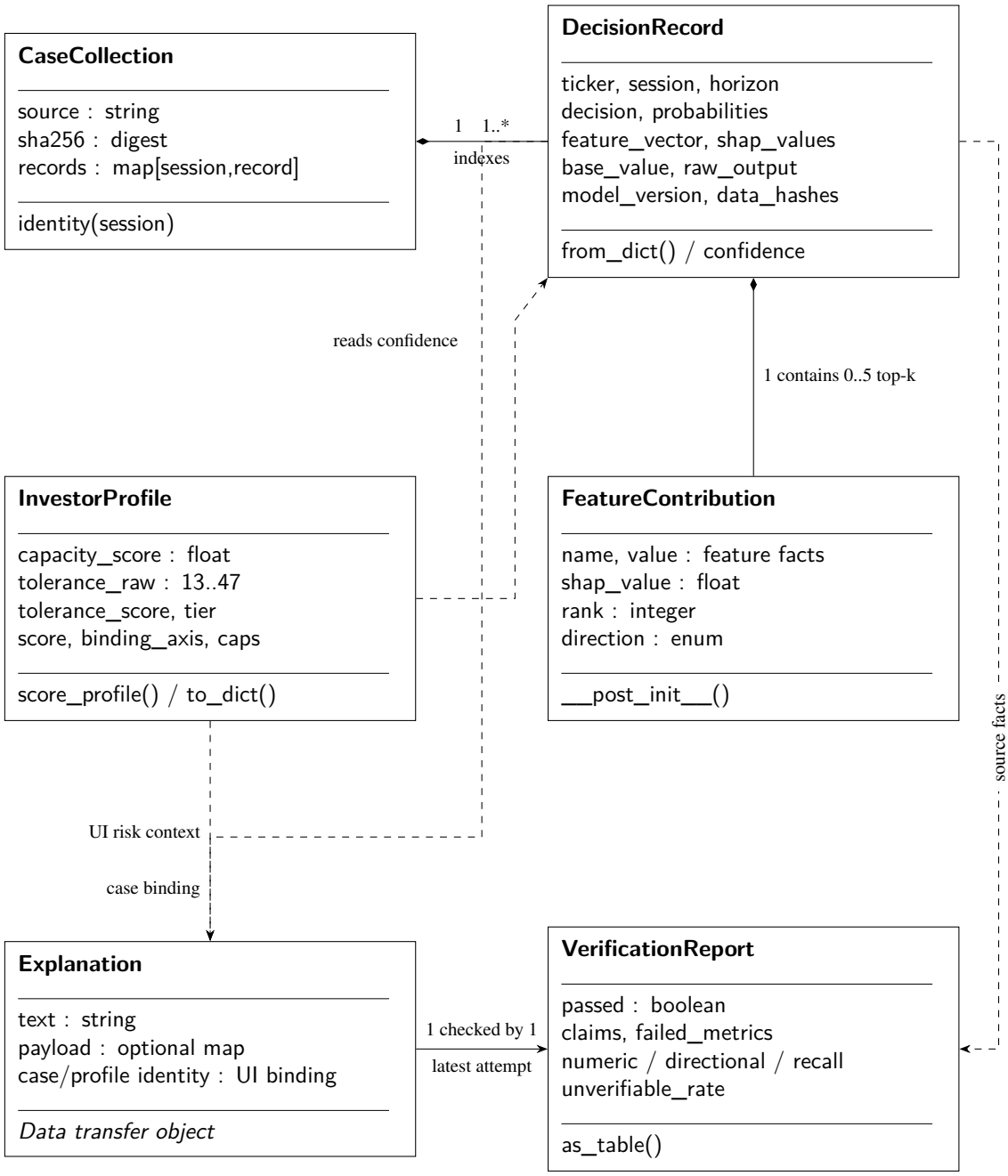


Figure 5.5: Implementation class model with attributes, operations and multiplicities.

SQL database. Relationship labels and cardinalities express how the actual loader and checker bind their data.

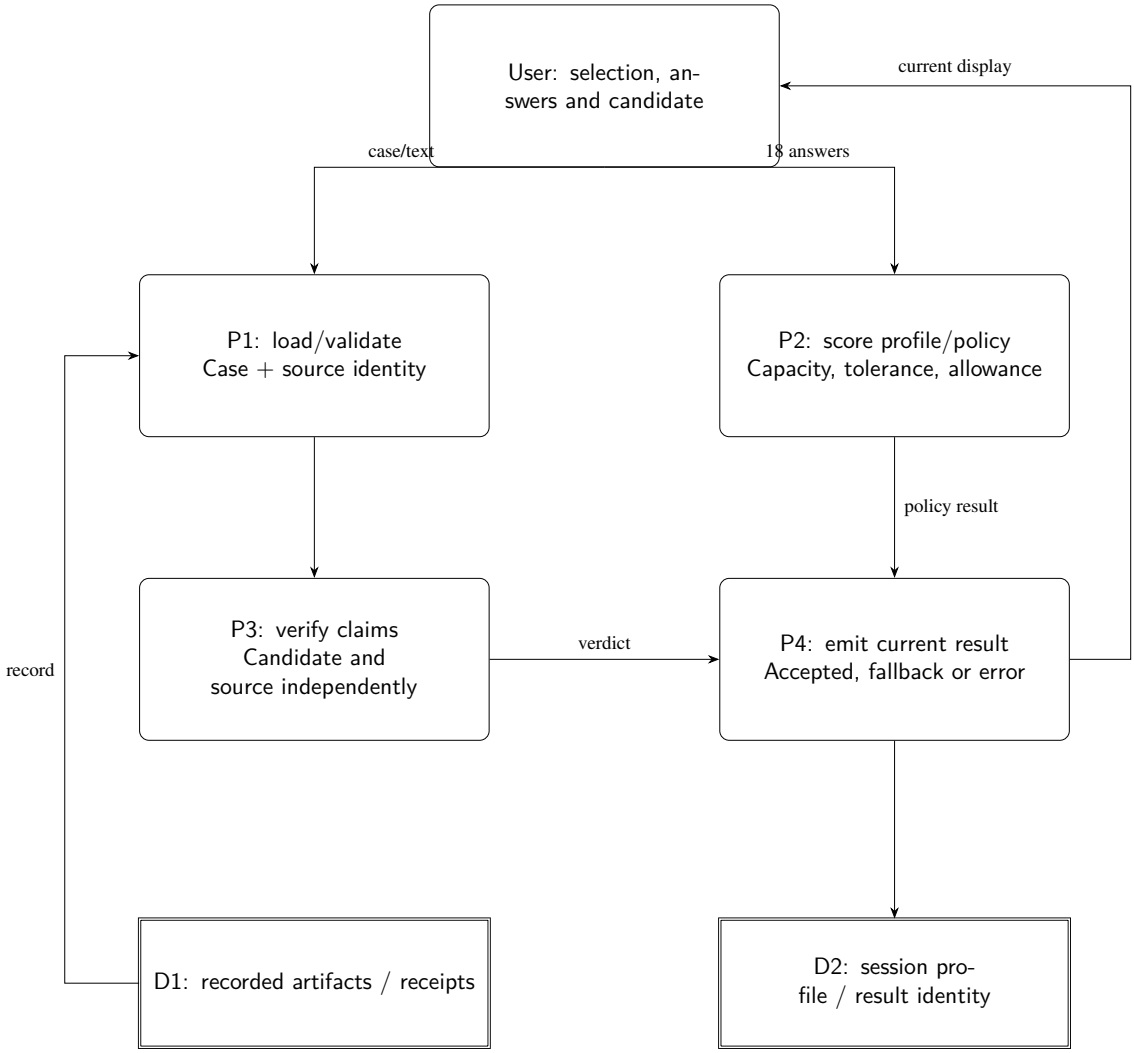
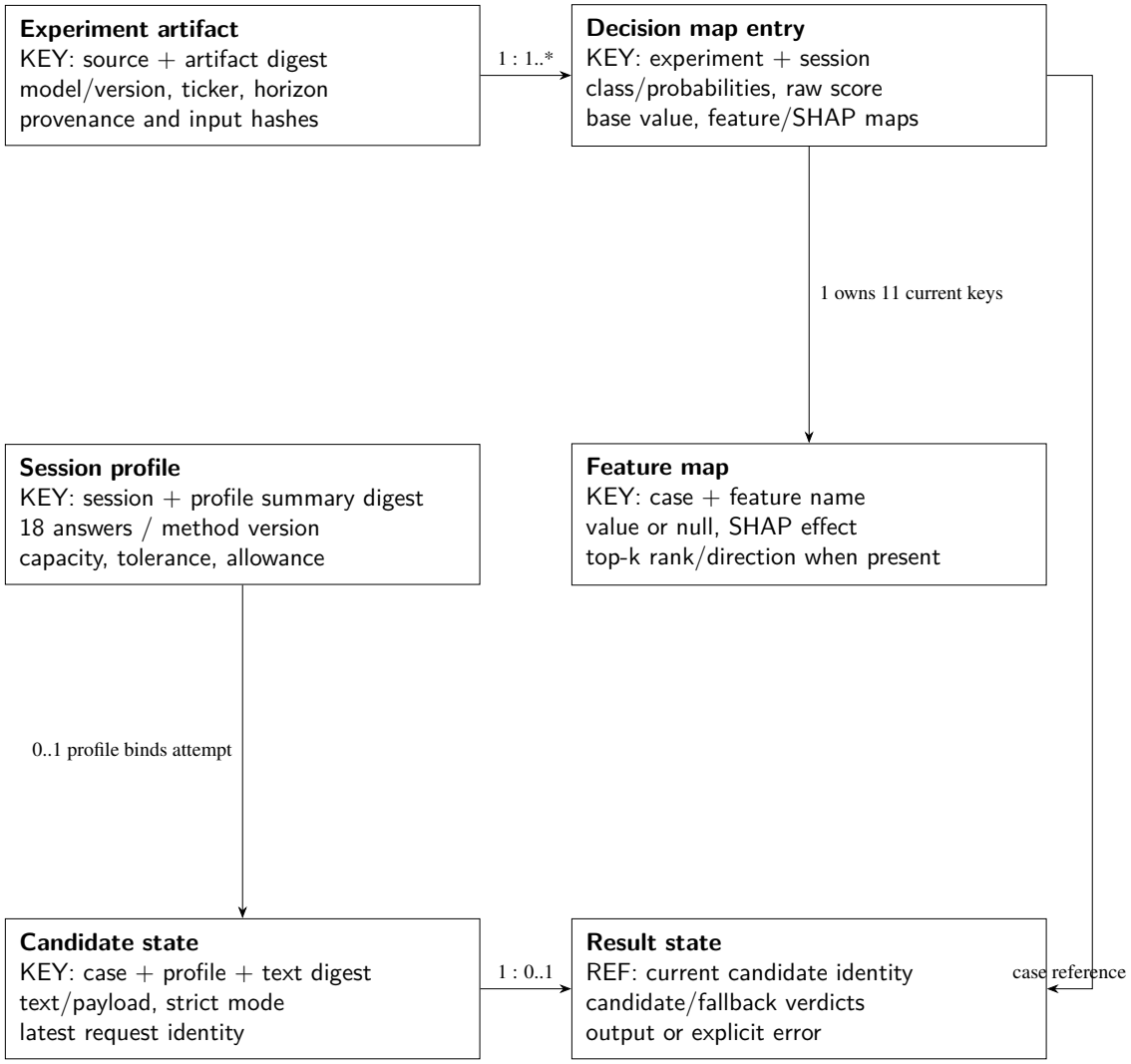


Figure 5.6: Data flow through the current checking and policy path.

5.5 Behavioral Design

The sequence starts with the actual Check explanation click and uses alternative fragments for acceptance and recovery. The activity includes invalid input and fallback-verification failure. The state model includes initialization, stale results, errors and session termination.



Keys describe actual file/maps and UI identity contracts. This is a schema model, not an implemented relational database.

Figure 5.7: Artifact/session schema with keys and relationship cardinalities.

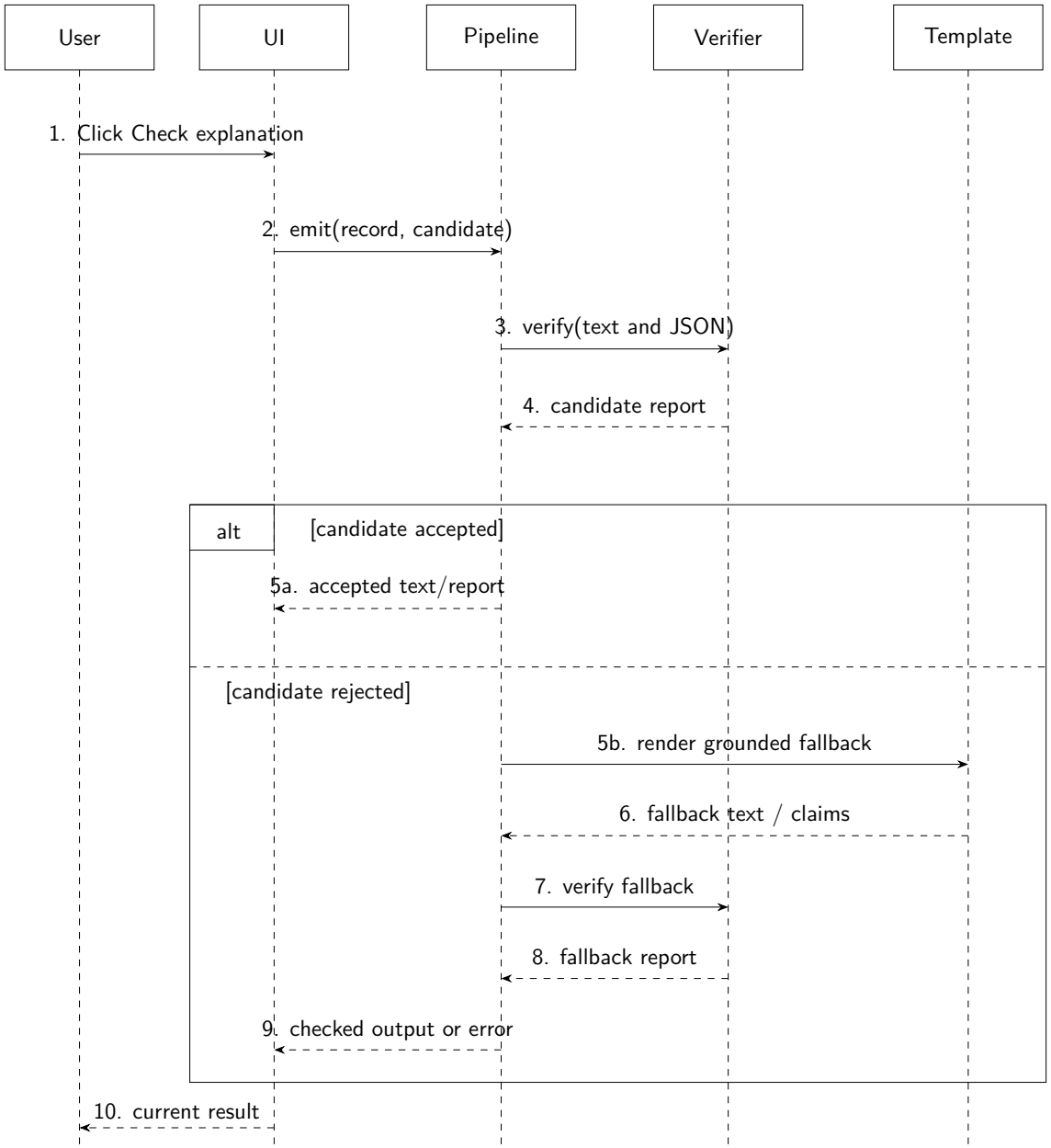


Figure 5.8: Explanation-check sequence with explicit user trigger and alternative outcomes.

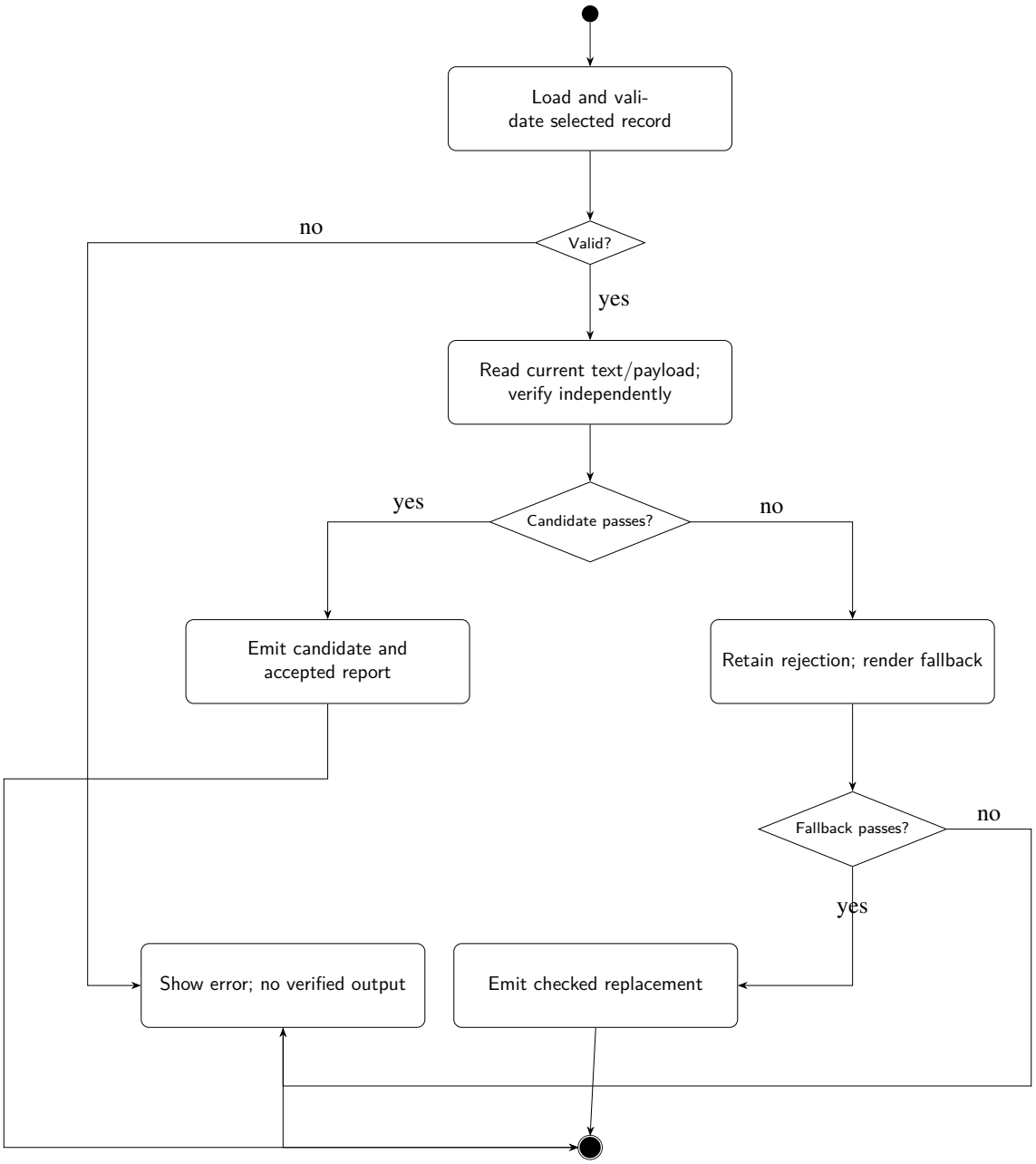


Figure 5.9: Checking activity, including verified fallback and error termination.

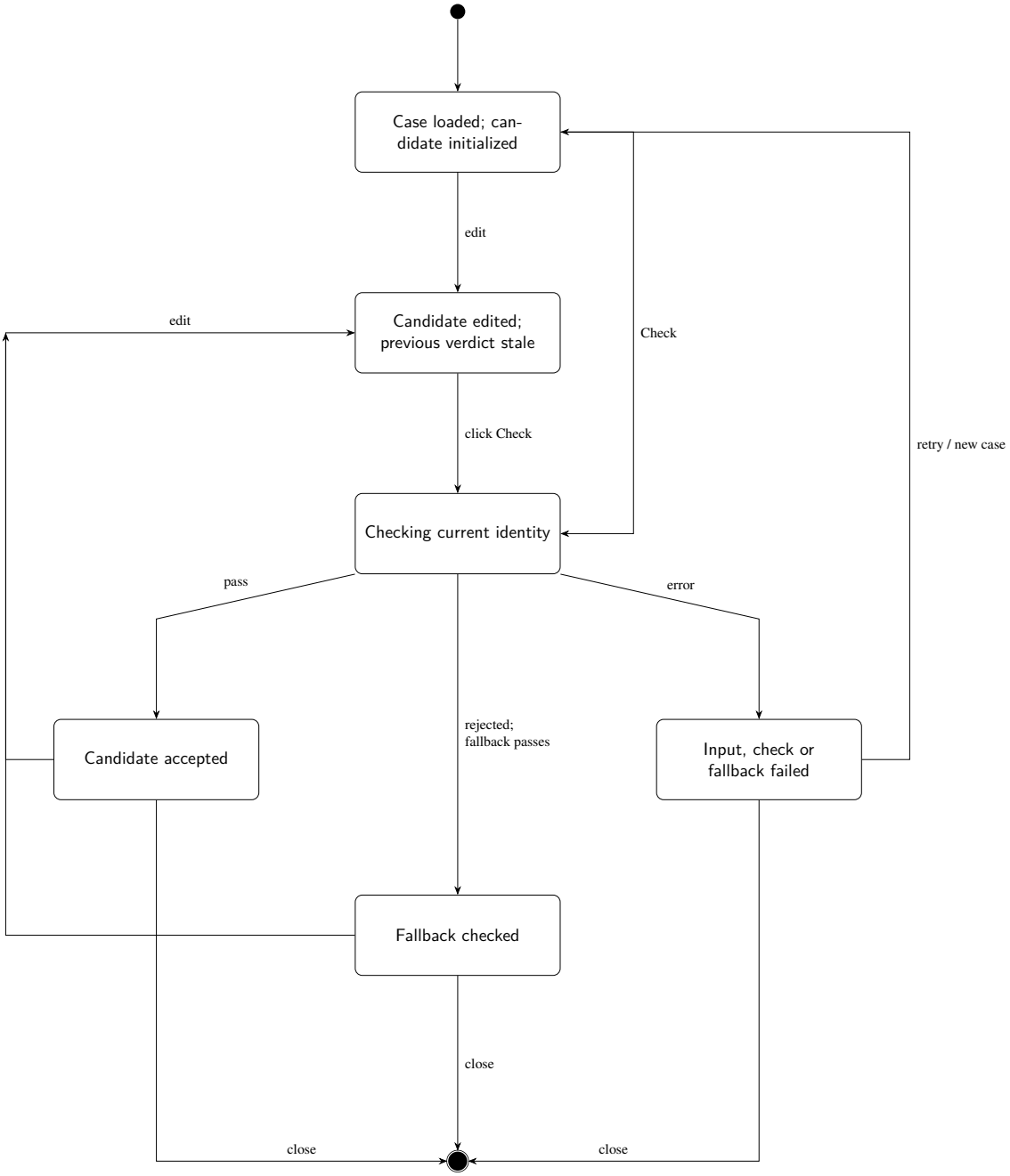


Figure 5.10: Case/candidate verification state transitions.

5.6 Deployment and Design Evolution

The localhost browser communicates with Streamlit. Core modules read registered artifacts and retain profile/candidate state in memory. CI produces downloadable review artifacts; it does not host the application. The revised deployment model includes the core-to-artifact link. The proposal's broad product design is retained, while the imple-

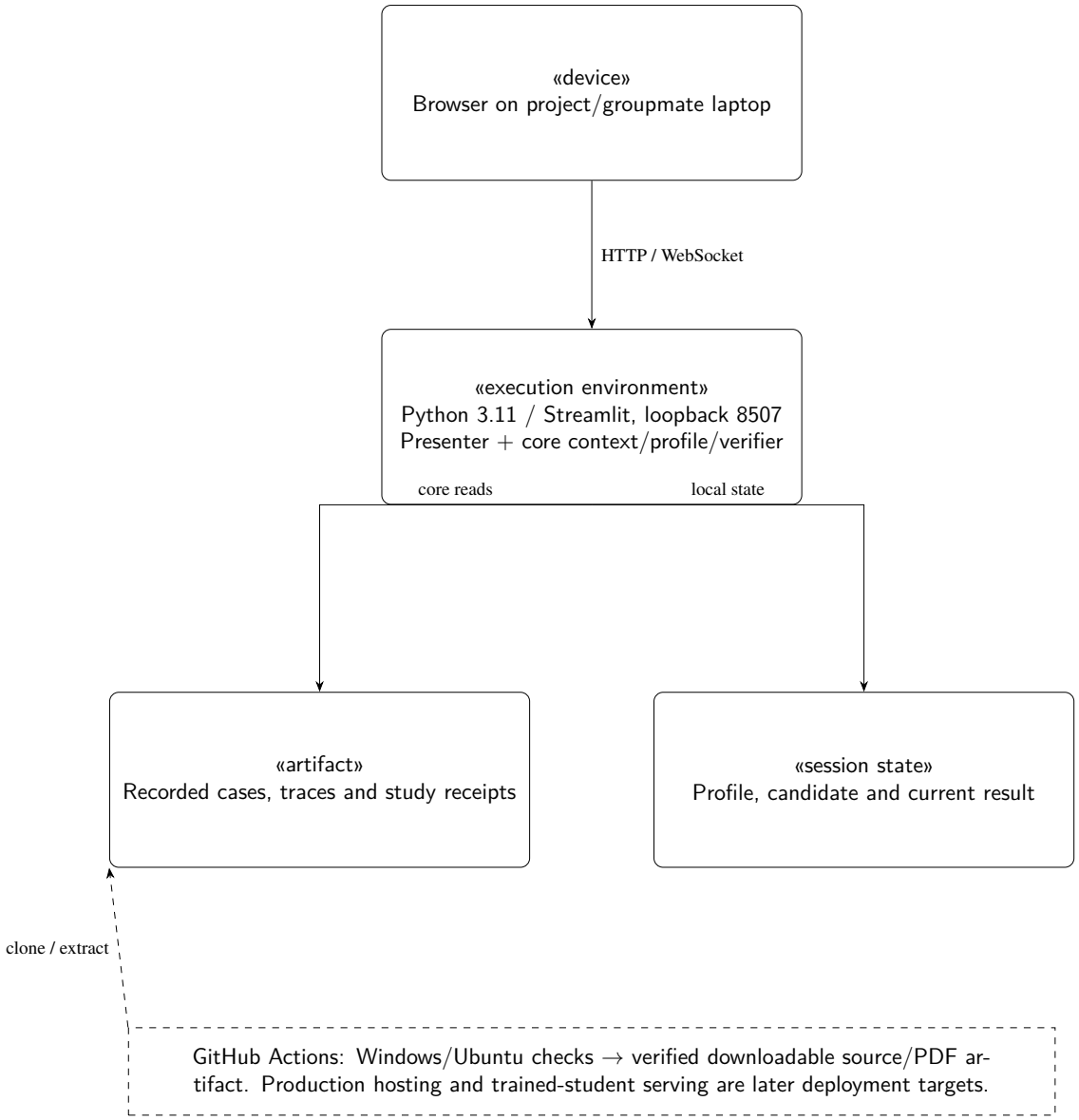


Figure 5.11: Current localhost deployment and artifact-delivery boundary.

mented slice now has complete evidence, independent prose/JSON checks, a full scored profile and uncertainty gating. The prior risk-dependent sentiment blend and isotonic outputs remain historical comparisons. Broader conversation, social divergence and production service work are explicitly scheduled.

5.7 Algorithms and Contracts

Record validation precedes display. Checking retains candidate errors, constructs fallback and verifies it independently. Temperature fitting admits only earlier-matured calibration labels. Profile scoring validates every answer, calculates weighted capacity/caps and published item totals, reconciles restrictive limits and evaluates confidence without changing direction. Each boundary carries identity and failure conditions used by its tests.

Chapter 6

Iteration 1: Implementation and Scientific Execution

6.1 Implemented Pipeline

The pipeline validates versioned price/news snapshots, maps timestamps to exchange sessions, builds causal indicators and news aggregates, and trains/evaluates a fixed XGBoost configuration. Each decision record retains identity, probabilities, complete feature/SHAP maps, raw score, model version and input digests. The interface reads those records; the verifier and risk calculations execute live.

6.2 Data Interval and Source Decision

Table 6.1: Consumed historical data and final evaluation interval.

Input	Interval/count	Policy
NVDA daily prices	2 Jan 2018–29 Dec 2023; 1,509 sessions	SIP split-adjusted snapshot; cash dividends not adjusted
Alpaca/Benzinga news	2018–2023; 6,768 retained rows, 5,748 distinct headlines	Availability is later creation/update time
Model evaluation	29 Mar–21 Dec 2023; 186 decisions per mode	Five observed-session outcomes

Calibration	181 available earlier outcomes	Five outcomes crossing evaluation start excluded

FNSPID remains a preserved historical comparator. Its original audit found that 99.78% of the selected NVDA rows carried midnight-only timestamps, with only 26 genuinely intraday-timed rows; AAPL/TSLA also had material year gaps. Those limitations cannot support a broad precise-publication alignment claim. Alpaca/Benzinga was used for the new timestamped study, consistent with the original repository's documented recovery. The scope remains NVDA until multi-company coverage is audited. Current-vintage data is not claimed as a byte-identical September reproduction.

6.3 Temporal Controls and Terminology

The target spans five observed trading sessions. Purging excludes a training label whose actual endpoint reaches the evaluation start. The configured five-day embargo is an additional calendar-day gap: these are different rules. Replacing endpoint logic with calendar subtraction retained 18 invalid endpoints on the boundary fixture, while the corrected purge retained zero. The explanation of this distinction resolves the proposal/report terminology difference rather than treating the two terms as synonyms.

The no-purge arm retains an additional gap and may show no detectable performance inflation when only a few boundary outcomes are affected. The same-bar branch remains a lag ablation under the strict-prior join, not a guaranteed future leak. A centred window explicitly consumes future observations and is the permanent canary. The shuffled split breaks chronological evaluation. Each arm has a different fault geometry; not every deliberate variation must inflate the measured score.

6.4 Baseline and Leakage Results

Table 6.2: Actual five-session walk-forward/control results.

Arm	Macro-F1	Lift over random	Interpretation
Clean	0.2843	-0.0444	Below random comparator
Shuffled folds	0.4457	+0.1142	Chronological fault shows signal

No purge	0.3022	-0.0267	No measured inflation under this geometry
Centred Bollinger	0.5390	+0.2104	Explicit future-information canary
Centred SMA200	0.3333	+0.0046	No detectable uplift for this variation
Same-bar text	0.2651	-0.0636	Lag ablation, not evidence of future access

The centred-Bollinger lift interval is [+0.1698,+0.2421]. Results use fixed settings, shared row masks and six folds. Weak clean performance is retained; the experiment does not establish a useful forecasting strategy.

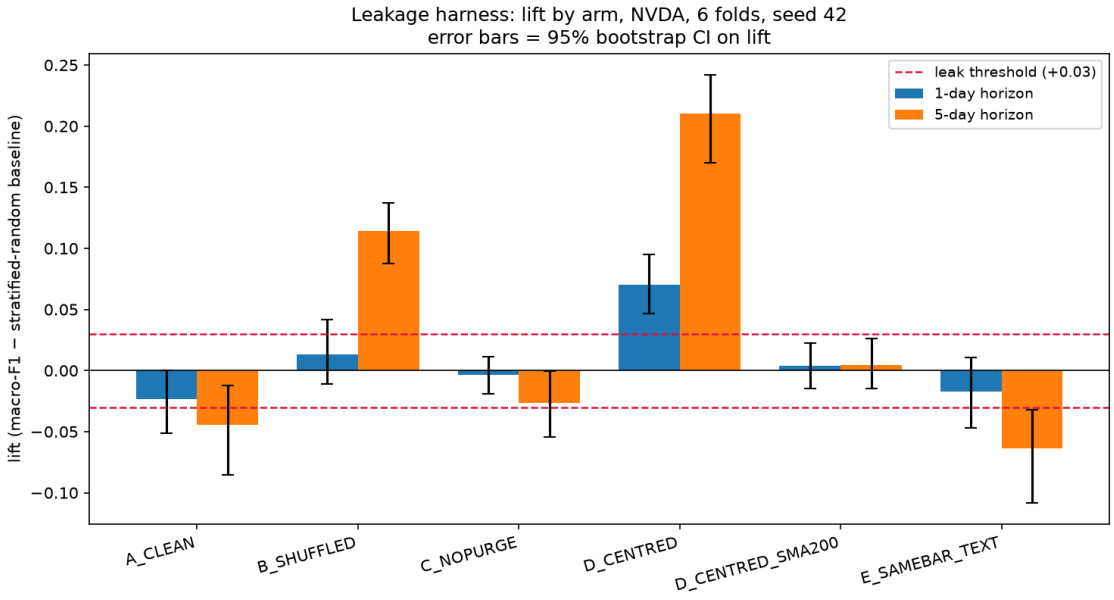


Figure 6.1: Retained leakage-arm lift and intervals; centred and shuffled faults fire, while other variations do not produce guaranteed uplift.

6.5 Calibration Diagnosis and Correction

Independent isotonic maps force Buy because its pre-normalization minimum 0.5344828 exceeds Hold maximum 0.2500000 and Sell maximum 0.3139535. Normalizing by a common positive sum preserves that ordering. We retain those records as the diagnosed

failure and construct a new global-temperature experiment from the same immutable model bytes, scaler, margins and input rows.

Table 6.3: Held-out calibration behavior on the same 186 evaluation cases.

Method	Buy/Hold/Sell	Macro-F1	Brier
Frozen raw	40 / 51 / 95	0.3182	0.7929
Frozen isotonic	186 / 0 / 0	0.2238	0.6105
Frozen temperature	40 / 51 / 95	0.3182	0.6667
Refit raw	37 / 50 / 99	0.3116	0.7760
Refit isotonic	186 / 0 / 0	0.2238	0.6078
Refit temperature	37 / 50 / 99	0.3116	0.6667

The single temperature is selected by negative log likelihood on the 181 earlier calibration outcomes; evaluation labels do not fit it. Its optimum reaches the configured upper bound of 1,000, making probabilities nearly uniform. Rank preservation removes the artificial all-Buy collapse, but does not add predictive information. The interface therefore exposes the diagnostic class and withholds a directional recommendation at insufficient confidence. Brier and classification metrics are reported together; the correction is not marketed as a profitable or well-calibrated predictor.

Native XGBoost replay reproduces the retained margins. TreeSHAP is recalculated for each newly selected raw class and checked for additivity. Both numeric and complete-guided explanations are checked on all newly generated records. The new model identity keeps the original classifier and calibration findings traceable.

6.6 Complete Risk Profiling and Policy

Five capacity factors use retained weights: horizon/buffer 0.30 each, income stability 0.20, dependants/age 0.10 each. Protective caps handle no buffer, short horizon and unstable income with thin reserves. Thirteen Grable–Lytton item scores produce $G \in [13,47]$ and normalized tolerance $T_r = (G-13)/34$. The risk allowance is $R = \min(C, T_r)$; the more restrictive capacity/tolerance tier is displayed.

The primary policy requires confidence $\tau(R) = 0.80 - 0.30R$ for a directional suggestion. A higher allowance lowers that threshold but cannot change Buy into Sell or Sell into Buy. Weak evidence remains Observe; an existing Hold class remains non-directional. The old $C = wS + (1-w)T$ blend remains a historical baseline with explicit weighting, not the meaning of the current risk score. The published scoring is implemented; local

paraphrases and the confidence policy require further suitability evaluation.

6.7 Explanation Checking and Teacher Construction

The checker validates actual visible prose independently of JSON, including probabilities, values, feature names, attribution direction/rank, coverage and maintained meanings. Known contradictions fail before rate-based metrics can dilute them. Rejection preserves the failed report and triggers an independently checked deterministic fallback.

The teacher construction pilot contains six actual recorded explanations: four training and two validation exports. Factual JSON is assembled from the source record and is distinct from generated prose. Grouped dates keep test sessions outside the pilot. This establishes example construction and bounded checking, not trained-student efficacy or participant understanding. The larger prompted/student/teacher evaluation remains a later milestone.

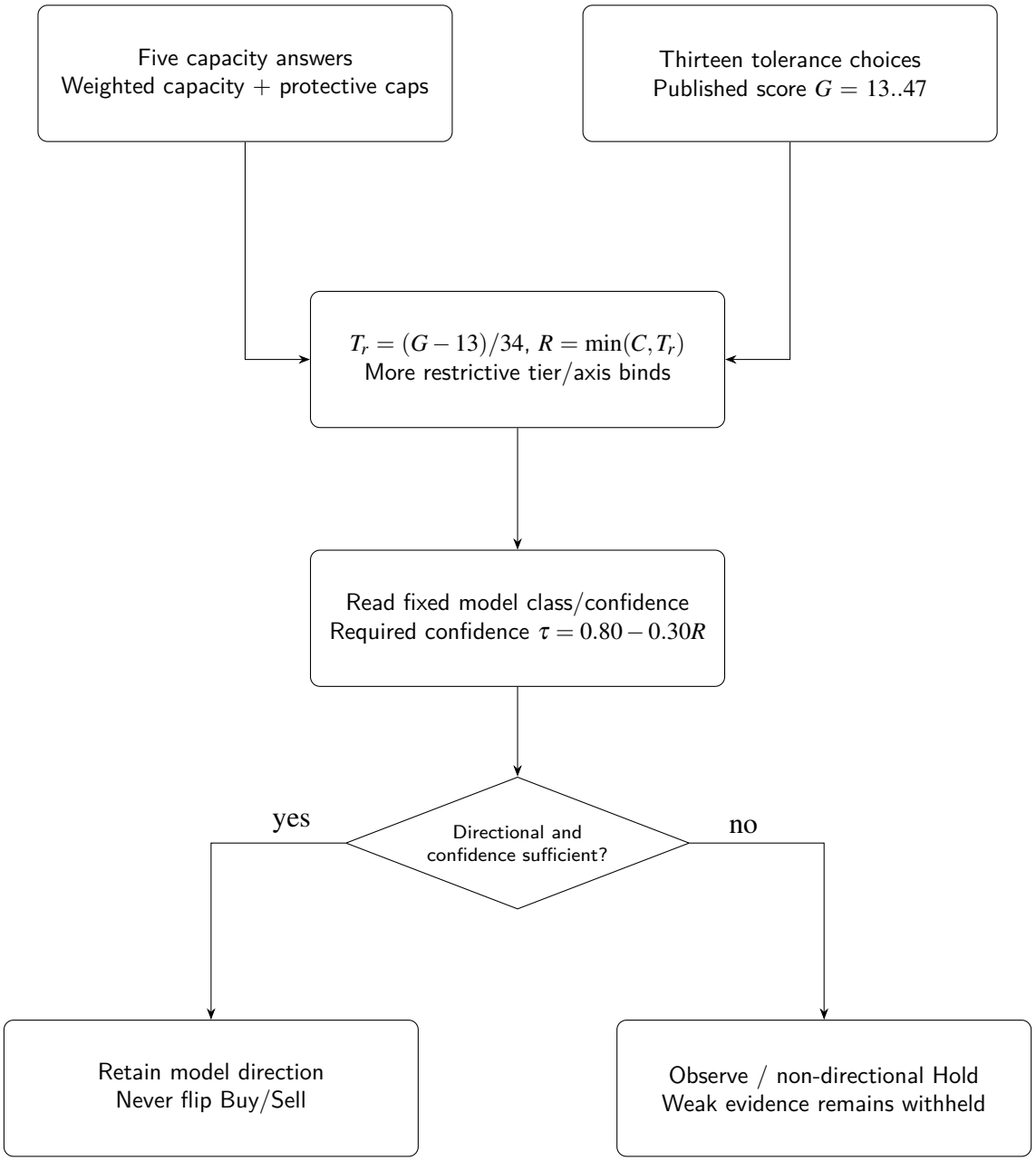


Figure 6.2: Risk capacity/tolerance calculation and direction-preserving confidence policy.

Chapter 7

Interface Design and Setup Manual

7.1 Current and Planned Interface

The implemented interface has Overview, Walkthrough, Experiments and Design. The four walkthrough steps are Decision, Evidence, Risk profile and Explanation check. Only one workspace/step executes at a time. Figure 7.1 shows the implemented slice; Figure 7.2 shows the planned end-user progression without presenting it as completed software.

7.2 Local Installation

Install Git and Python 3.11, clone the repository, create a virtual environment and install `code/requirements-demo-runtime.txt`. Restore reviewed checkout bytes with `tools/restore_ci_checkout.py`. Start the app from the repository root:

```
python -m streamlit run code/app/streamlit_app.py
  --server.address=127.0.0.1 --server.port=8507
```

The command should be entered on one line. Open http://127.0.0.1:8507 and keep the terminal open. No GPU, model download or API key is needed for recorded viewing. The prepared project laptop can use `tools/start_guided_demo.ps1`; a different machine should follow the portable setup. An occupied port or missing runtime produces an error; use an available port/environment rather than stopping an unrelated process.

7.3 Live Route and Observable Results

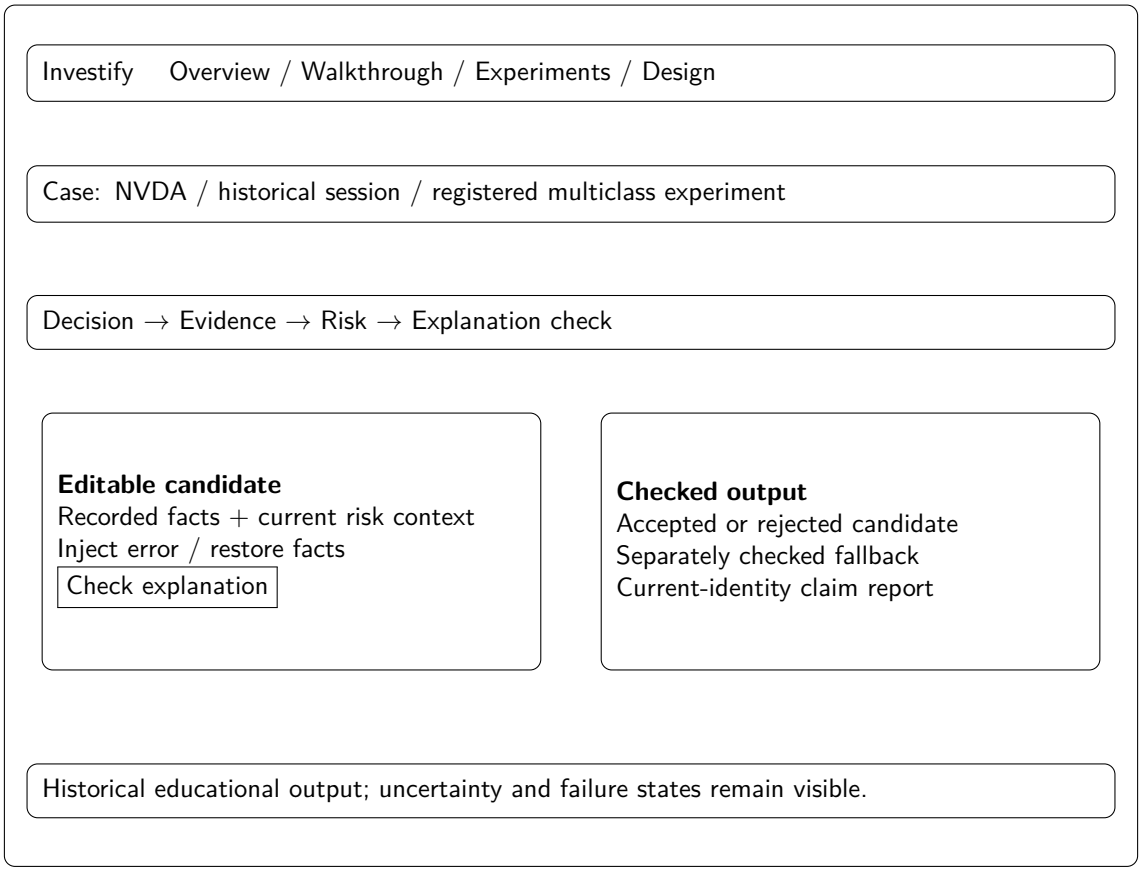
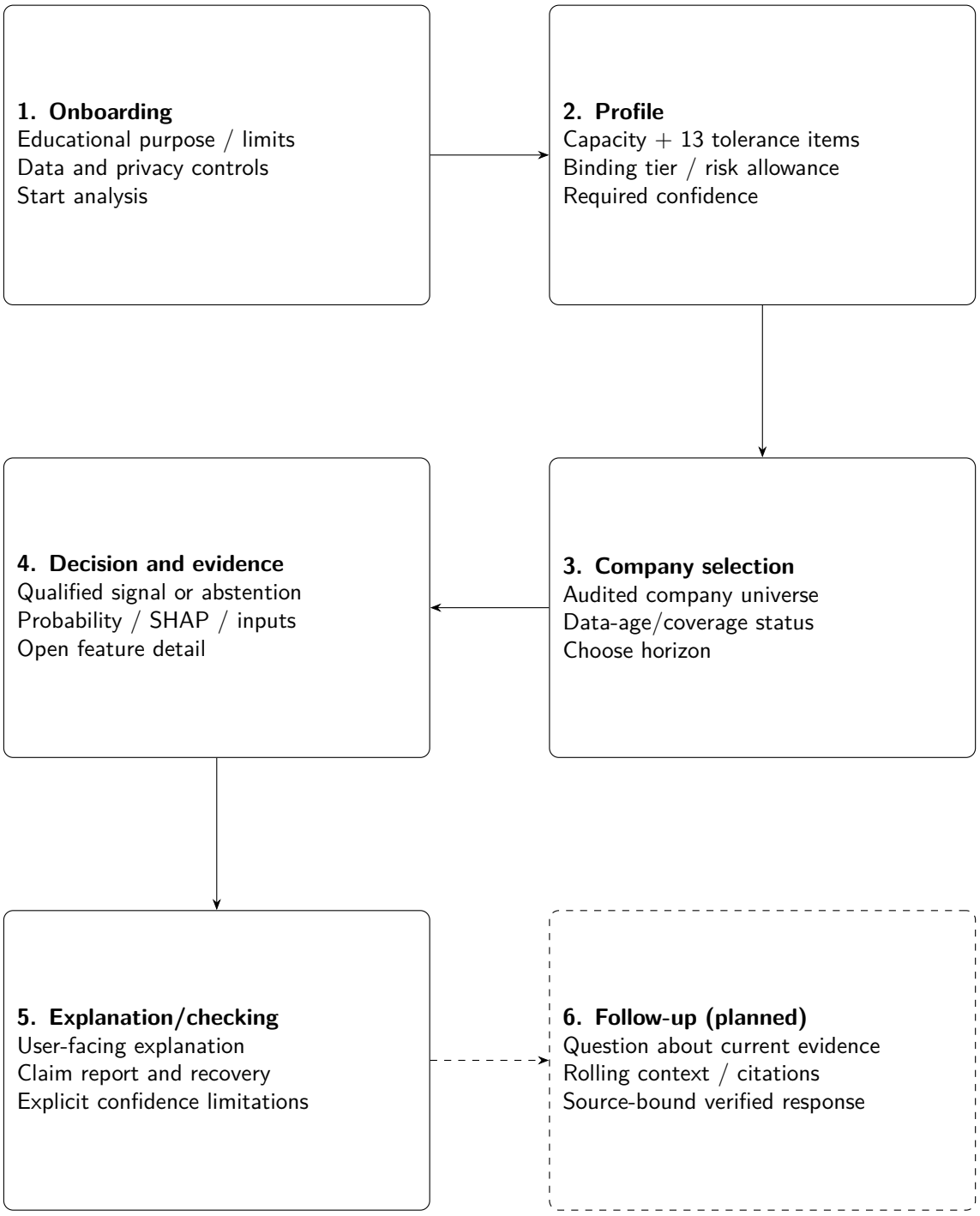


Figure 7.1: Current explanation-check interface wireframe.



Wireframes describe the product journey. Dashed follow-up and multi-company/service integrations are future scope; actual screenshots are shown separately.

Figure 7.2: Planned onboarding, profile, company, evidence and follow-up interfaces.

Table 7.1: Current user actions and observable responses.

Step	Action	Response
Decision	Select multiclass-calibrated frozen NVDA case, 21 Dec 2023	Raw class and probabilities; insufficient-confidence state
Evidence	Select technical/news feature	Recorded value, meaning and raw-score effect
Risk	Complete eighteen answers or choose labelled example	Published tolerance total, capacity limit, allowance and confidence gate
Explanation	Inject 1.00%, then Check explanation	Rejection and separately checked factual fallback
Recovery	Edit candidate or change case	Obsolete verification cleared
Studies	Select retained experiment	Source-bound results and limitations, without inference

Both example profiles abstain on near-uniform real evidence; a higher allowance lowers the required confidence rather than reversing the market class. Controlled policy tests exercise qualifying Buy/Sell cases separately. The preserved isotonic choices are labelled diagnosed failures and are not eligible for directional suggestions.

7.4 Failure Handling

Missing records, invalid profiles, inconsistent study identities and checker errors withhold substitute outputs. A failed fallback produces an error. The optional generator is called only by explicit action, and its absence is labelled. Brokerage accounts, orders and trained-student serving remain outside this prototype.

7.5 Actual Interface Captures

The revised captures show the decision/abstention, complete-profile result and rejection/fallback panels at readable scale. They are actual browser panels, not wireframes. Captures are produced after code verification and remain distinct from a physical projector check.



Figure 7.3: Actual revised decision panel: approximately 33.3% confidence and explicit abstention.

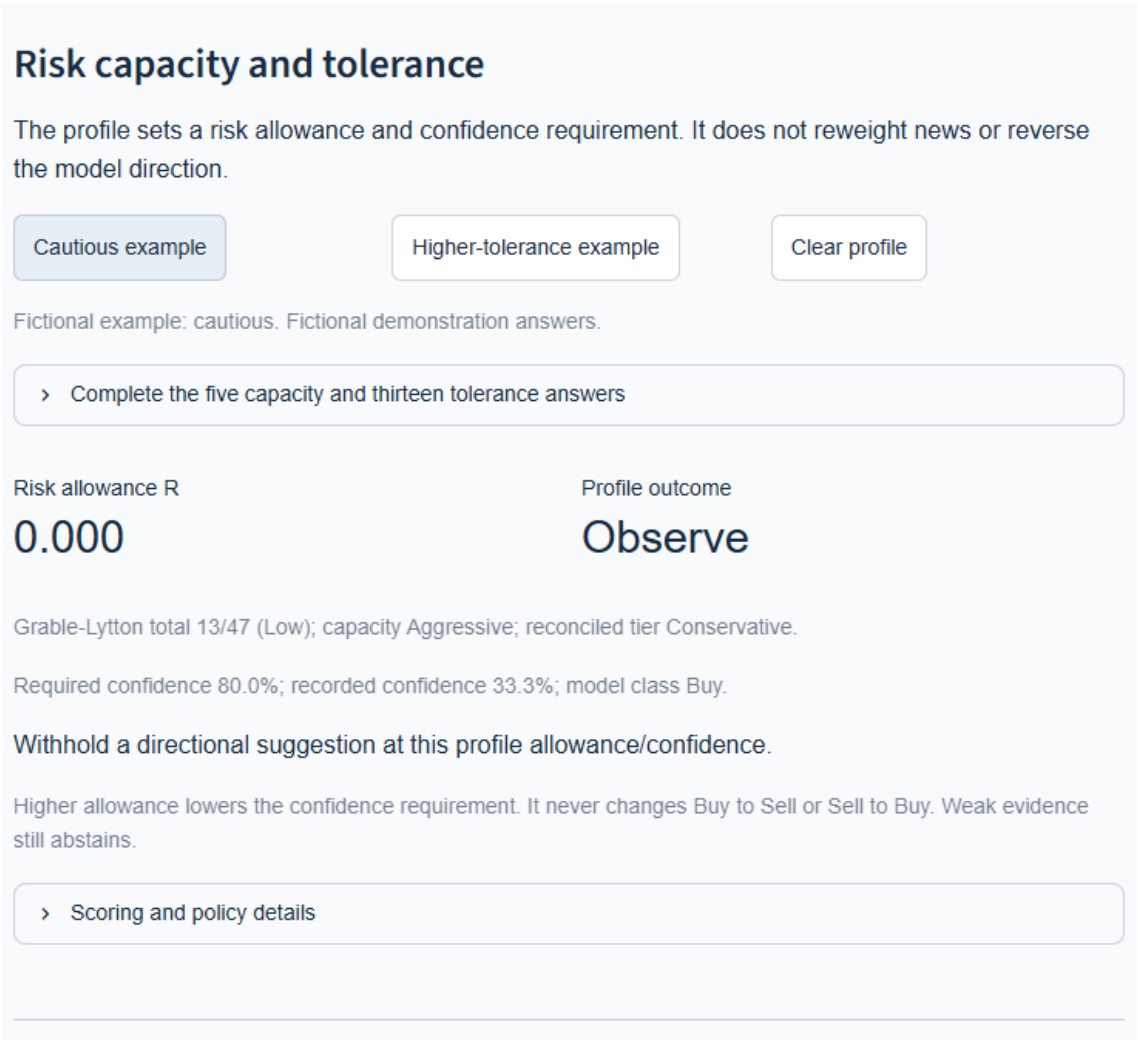


Figure 7.4: Actual complete-profile result: published total 13/47, allowance zero and an 80% confidence requirement.

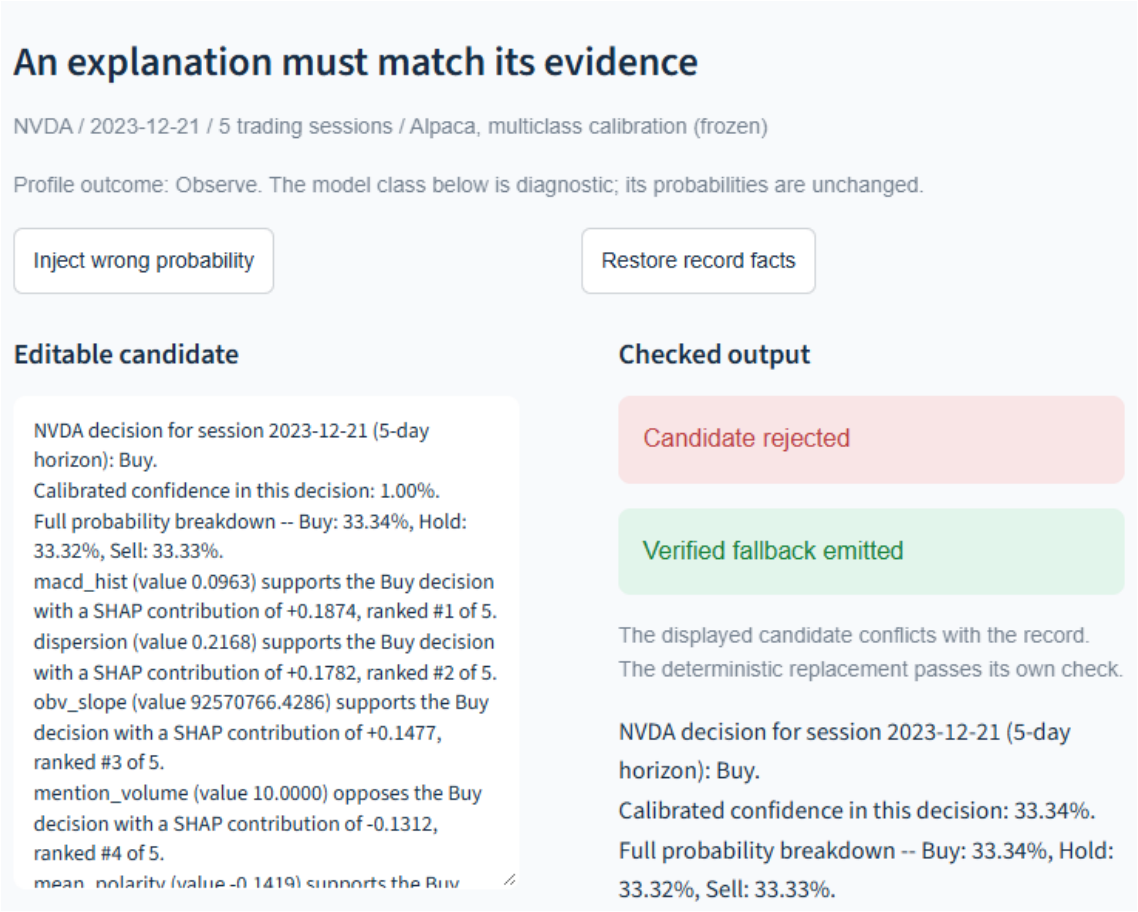


Figure 7.5: Actual visible 1.00% corruption rejected; independently checked fallback restores 33.34%.

Chapter 8

Testing and Early Validation

8.1 Strategy and Test Levels

Unit tests exercise validation, formulas and hard rejection gates. Module tests combine records, evidence, profiles and emitted explanations. Streamlit AppTest exercises actual widgets; a localhost HTTP exercise validates the portable pipeline. Source/receipt guards and clean extraction verify artifact reproducibility. Scientific execution and classifier sanity probes are separate from software fixtures.

8.2 Evaluation Metrics Matrix

Table 8.1: Metrics, evaluation population and measured/planned status.

Area	Measures	Evaluation population/protocol	Status
Prediction	Macro-F1, per-class precision/recall, baseline lift	NVDA chronological folds; random and linear comparators	Measured; weak signal
Calibration	Brier, class distribution, negative log likelihood	Earlier calibration labels; same 186 evaluation cases	Measured; abstention required
Leakage	Control lift/intervals, boundary/fault detection	Six arms, two horizons, fixed folds/settings	Measured; labelled canary fires

Explanation	Numeric/directional fidelity, recall, unverifiable rate; rejection/fallback	Saved cases, deliberate corruptions and construction pilot	Bounded behavior measured; larger generation study later
Social sentiment	Macro-F1 agreement, per-class recall, coverage	Same 24 held-out target-equity references	Exploratory comparison measured
User understanding	Task completion/time, interpretation errors, top-driver identification	Planned participant tasks and expert review	Later evaluation
System	Latency, memory, throughput, parity and setup	Specified target environment and clean-host paths	Setup/path checks measured; target NFRs later

8.3 Current Software Checks

The current revision adds calibration-availability/rank tests and full-profile scoring/policy tests. Existing corruption, record, transport and interface regressions remain. Named input/expected/actual cases are listed in Appendix C; final scoped receipts record the exact executed counts. Historical counts are not repeated as if they were one current suite. The hosted matrix proves its committed snapshot; new local changes require their own checks before publication.

8.4 Social Study and Diagnostic Probes

Table 8.2: Social study population, split and original-classifier outcomes.

Item	Value	Interpretation
Acquired / usable / excluded	200 / 117 / 83	Exclusions are retained with reasons
Usable reference classes	31 positive / 67 neutral / 19 negative	Imbalanced target-equity references
Development / held out	93 / 24	Frozen chronological selection

Held-out classes	7 positive / 15 neutral / 2 negative	Negative recall moves in 50-point steps
CryptoBERT	Macro-F1 0.2564; exact 15/24	All held-out predictions neutral
Cardiff	Macro-F1 0.6016; exact 16/24	Exploratory leader, not final deployment selection

The reference-construction method is disclosed in Appendix G. Scores measure agreement with that reference, not independent human-validated accuracy or NVDA return prediction.

A new offline sanity check uses five authored crypto-text probes, kept separate from the frozen equity benchmark. The original CryptoBERT checkpoint returns both positive and negative classes: the two intended-positive and two intended-negative probes match their diagnostic direction. The nominal neutral probe returns negative. This rules out an invariant neutral output on every input, while showing the probes are not a validated accuracy set. We retain the original 24-post result and do not relabel the benchmark to improve it.

8.5 Failure and Recovery Evidence

The demonstration pairs correct factual text with a deliberately incorrect visible probability beside correct JSON. The candidate fails and the checked fallback retains the authoritative probability. Edits/case changes remove stale verification. Missing records, partial model outputs and invalid profiles withhold replacements. Temperature calibration preserves raw class order; actual near-uniform confidence abstains under both profile extremes. Controlled 0.70-confidence Buy/Sell cases demonstrate that allowance changes abstention without reversing direction or mutating probabilities.

8.6 Defects Corrected

Corrections address payload-over-prose acceptance, missing/non-top-k evidence checks, retained rejection reporting, stale state, label-endpoint purging, calibration-outcome availability, and platform-specific provenance-path handling. The present revision adds multiclass diagnosis/abstention and replaces risk-dependent sentiment weighting in the primary interface. Old model/data/label artifacts remain separately identified.

8.7 Validation Boundary

The live module and engineered failures are observable and tested. Predictor usefulness, unrestricted generated-text quality, population-specific suitability and participant comprehension are separate evaluation goals. Physical rehearsal/projector checks and formal meeting/signature records remain team activities.

Chapter 9

Project Management and Development Evidence

9.1 Work Delivered

The team delivered the historical data/evaluation path, complete evidence context, engine/attribution outputs, live verification/recovery, profile interface and source/PDF delivery process. The present revision corrects calibration presentation, profile semantics, requirements detail and report diagrams. New scientific outputs retain separate identities.

9.2 Individual Responsibilities and Contributions

The work division follows the proposal and the module preparation delivered for this evaluation. Each member contributed an equal one-third share to module delivery, integration preparation and defence responsibility. Members continue studying and practising their code as part of preparation.

Table 9.1: Delivered member work and continuing responsibilities.

Member	Delivered Mid-1 part	Continuing work
Ahmed Javaid	Temporal data/session alignment, label/purge and leakage evidence; CI/delivery and reproduction preparation	Broader data audit, serving parity and deployment

Rafay Ul Haq	Interface/evidence journey, capacity/tolerance profiling, sentiment-study and failure/recovery preparation	Social integration, degraded operation and acceptance evaluation
Soban Ahmad	Decision/calibration/SHAP evidence, claim verifier and fallback, teacher/student pipeline preparation	Prediction/calibration analysis, student inference and QLoRA comparison

9.3 Individual Defence and Integration

Ahmed presents the temporal harness and testing/delivery evidence; Rafay presents the user, risk and sentiment path; Soban presents model/attribution and explanation checking. Each has four primary slides and five planned speaking minutes including the shared demonstration. The team guide identifies the actual files/functions, named tests and questions to practise. All three also review the full integrated path.

9.4 Repository Evidence

The repository is https://github.com/Ahmed-Javaid/Investify-FYP-2027. Main commit `70bf0e3` is the previously published baseline. Earlier scientific and source snapshots remain in history. New report/code revisions have their own local receipts until committed/published; earlier hosted success is not presented as proof of uncommitted changes.

9.5 Continuous Integration and Delivery

Two Python 3.11 jobs execute portable and Streamlit scopes on Windows and Ubuntu. A fifth delivery job requires all four passing, clean, same-commit receipts, checks source/log identities, packages reviewed PDFs/source, verifies extraction and uploads the artifact. Push, pull request and manual dispatch trigger the workflow; artifacts are retained for fourteen days. Continuous delivery means tested artifacts, while production hosting remains later work.

The retained five-job successful run is https://github.com/Ahmed-Javaid/Investify-FYP-2027/actions/runs/38428835203. The earlier successful artifact was downloaded and independently checked against the server digest and delivered-file

hashes. The workflow exercises recorded data, real UI/HTTP paths and packaging; it does not train students, acquire current news or compile/render PDFs automatically.

9.6 Milestones and Recovery

The dated plan in Table 3.3 assigns owners to the remaining integration and evaluation. Corpus precision, weak prediction and calibration collapse are explicit findings with concrete recovery actions. The current response prevents misleading directional output without manufacturing a positive held-out result.

9.7 Privacy and Responsible Development

The local interface keeps profile answers in session memory and provides no brokerage access. Credentials and model caches are excluded from handover artifacts. Reusable components and development assistance are disclosed in Appendix G. The team accepts responsibility for source review, integration, factual reporting and independent defence.

Chapter 10

Remaining Development Plan

10.1 Remaining Scope

The full product requires audited multi-company inputs, news/social divergence, improved prediction/calibration, population-specific profile interpretation, evaluated generated explanations, serving/deployment and participant acceptance. QLoRA is a later milestone; brokerage execution remains excluded.

10.2 Integration and Evaluation

The next analysis compares declared baselines and development-only model/calibration choices. We retain weak findings rather than selecting on held-out scores. Social deployment selection follows agreed adequacy criteria; the divergence contribution requires a predeclared ablation on integrated data. Student work begins with a verified original checkpoint, a scored prompted baseline and a larger filtered training set.

10.3 Operational Validation

Measure target latency, memory and throughput; run live/backtest parity with a pinned clock; inject unavailable/thin-history sources; and document degraded responses. Production hosting and external installation require an explicit target and reproducible service package. Hosted artifact delivery is already implemented and remains distinct from hosting.

10.4 Participant Evaluation

Later tasks ask participants to interpret a recommendation, its drivers, uncertainty and verifier report. Record completion, time, assistance and interpretation errors with consent. The published risk instrument motivates scoring; local wording and policy suitability still need review. Tool/fixture success is not participant acceptance.

10.5 Timeline and Final Deliverables

The dated ownership plan appears in Table 3.3 and Figure 10.1. Final deliverables include integrated source/service, reproducible acquisition/evaluation, audited explanation/profile behavior, external installation and participant records, report/presentation and handover instructions. Formal assessment dates remain distinct from internal engineering targets.



Internal engineering targets; formal assessment dates are separate.

Figure 10.1: Dated engineering sequence from Mid-1 correction to final handover.

Chapter 11

Conclusion

Investify delivers a historical evidence workflow, temporal evaluation controls and independently checked explanations. The tests and live failure path demonstrate implemented checking and recovery. The source audit documents why precise Alpaca/Benzinga timestamps were used alongside preserved FNSPID comparisons.

The calibration diagnosis is substantive: independent maps force Buy, while the new rank-preserving revision exposes nearly uniform confidence. The product response is abstention, not a fabricated useful prediction. Complete tolerance/capacity scoring now controls a direction-preserving confidence requirement. The team has delivered its assigned modules and a reproducible preparation package. Broader prediction, social integration, student adaptation, serving and participant acceptance remain the next dated milestones.

Appendix A

Proposal Action Register

The original supplied comments remain preserved in the source register. This table records the concrete response and the work that still requires evaluation.

Table A.1: Proposal actions and current implementation response.

ID	Requested change	Response / remaining evaluation
PF-01	Clear contribution	Harness and verifier separated from imported components; closest-work review remains
PF-02	LLM benefit	Rich evidence contract, teacher construction and planned comparison; participant benefit unmeasured
PF-03	Technical/sentiment explanation	All feature evidence and checked claims; integrated social evidence remains later
PF-04	Refined problem	Educational interpretation, temporal evaluation and factual consistency
PF-05	Model clarity	FinBERT news, evaluated social alternatives, frontier teacher construction and planned student
PF-06	Metrics	Explicit matrix in Table 8.1; measured/planned status distinguished
PF-07	Improved PoC	Live corruption/recovery, full profile, uncertainty gate and revised interface/design
PF-08	Transparency	Historical data, weak confidence, failed calibration, method/scope and fallback disclosed
PF-09	Market understanding	Timing, split/vintage, indicator meanings and attribution limits connected to design

Appendix B

Detailed Requirements Traceability

The versioned matrix maps requirements to actual source symbols and named regressions. It verifies implementation/test anchors; scientific quality and participant acceptance remain separate.

Table B.1: Functional requirement, implementation and regression traceability.

ID	Implementation	Named check
MID-FR-01	`DecisionRecord.from_dict`	`test_nonfinite_out_of_range_or_boolean_probabilities_are_rejected`
MID-FR-02	`build_context`	`test_all_real_records_keep_their_facts_and_validate`
MID-FR-03	`verify`	`test_correct_payload_cannot_hide_wrong_displayed_probability`
MID-FR-04	`verify`	`test_payload_cannot_supply_facts_or_notice_absent_from_displayed_text`
MID-FR-05	`emit`	`test_fallback_retains_the_rejected_candidate_report`
MID-FR-06	`verify`	`test_strict_mode_rejects_unparsed_text_and_preserves_clean_template`
MID-FR-07	`harness_input_report`	`test_empty_partition_directories_do_not_enable_a_rerun;test_leakage_view_labels_saved_results_and_limits_preflight_claims`

ID	Implementation	Named check
MID-FR-08	`LocalExplainer.explain;verify_runtime`	`test_unavailable_provider_fails_fast_with_explicit_fallback;test_all_shards_are_required_and_changed_bytes_fail_before_inference`
MID-FR-09	`verify_generated`	`test_wrong_reviewed_definitions_in_text_or_json_are_rejected`
MID-FR-10	`evaluate`	`test_plan_drift_is_rejected_before_any_provider_request`
MID-FR-11	`verify_generated`	`test_every_feature_has_a_value_magnitude_direction_and_target_in_both_sources`
MID-FR-12	`score_profile`	`test_incomplete_or_boolean_tolerance_answers_are_refused;test_capacity_caps_bind_despite_high_tolerance`
MID-FR-13	`build_comparator`	`test_changing_heldout_feature_values_cannot_change_scaling_statistics`
MID-FR-14	`apply_risk_policy`	`test_risk_allowance_changes_abstention_without_reversing_direction`
MID-FR-15	`validate_benchmark`	`test_one_ai_pass_has_null_agreement_and_explicit_reference_scope`
MID-FR-16	`evaluate_predictions`	`test_no_runs_or_one_candidate_never_manufactures_a_model_selection`
MID-FR-17	`run_inference`	`test_actual_loader_refuses_missing_weights_without_a_download`
MID-FR-18	`import_responses`	`test_actual_exports_keep_validation_out_of_training_and_disclose_hybrid_targets;test_actual_teacher_view_replays_examples_and_preserves_student_pending_state`

ID	Implementation	Named check
MID-FR-19	`fit_temperature;softmax_temperature`	`test_calibration_refuses_unavailable_labels;test_real_calibration_replay_preserves_all_raw_classes_and_probabilities`
MID-FR-20	`score_profile`	`test_thirteen_published_item_scores_have_exact_minimum_and_maximum`

The machine-readable matrix is `code/docs/requirements_traceability_v2.json`. The earlier eighteen-ID matrix remains preserved.

Appendix C

Individual Test Cases

Inputs and observations below distinguish real retained cases from explicit diagnostic fixtures. Final execution receipts identify the current source snapshot and test scopes.

Table C.1: Individual input, expected-result and actual-result test cases.

ID / case	Purpose	Input	Expected	Actual
TC-01	Valid NVDA case	2023-12-21, multi-class frozen	Valid source/date record; no replacement	Loaded; probabilities normalized
TC-02	Malformed input	Missing/invalid record	Error and no invented class	Error/refusal regression passed
TC-03	Displayed probability corruption	1.00% prose beside correct JSON	Reject candidate; checked fallback	Rejected; fallback accepted
TC-04	Structured contradiction	Correct prose, incorrect JSON claim	Independent payload rejection	Rejected in verifier regression
TC-05	Empty/omitted display claim	Missing decision or notice	No acceptance from payload alone	Rejected by hard coverage gate
TC-06	Unsupported strict sentence	Unparsed market prose	Strict rejection	Rejected; clean template accepted
TC-07	Wrong feature value/direction	Modified supported claim	Reject mismatch against record	Value and sign checks passed
TC-08	Fallback failure	Template fails verification	Error, no success claim	Failure path regression passed

ID / case	Purpose	Input	Expected	Actual
TC-09	Edited candidate	Edit after accepted verdict	Old verdict disappears	Widget invalidation passed
TC-10	Changed case	Select another session	Reset candidate/result identity	Widget case-change passed
TC-11	Incomplete profile	Only three tolerance answers	Refuse incomplete eighteen-answer form	ValueError; no profile
TC-12	Minimum tolerance	All published minimum choices	Total 13, allowance 0	13; R=0.000
TC-13	Maximum tolerance	Maximum choices, full capacity	Total 47, allowance 1	47; R=1.000
TC-14	Protective capacity cap	No buffer, high tolerance	Capacity binds below 0.4	Conservative; cap binds
TC-15	Direction preservation	Controlled Buy/Sell confidence 0.70	Low R abstains; high R retains direction	Low: Observe; high: original class
TC-16	Real uncertainty	Near-uniform corrected records	No unsupported directional suggestion	Hold remains neutral; others Observe
TC-17	Calibration availability	Outcome ends at evaluation start	Refuse fit	ValueError before fit
TC-18	Multiclass replay	186 margin vectors per mode	Same raw argmax; sum 1	All rows preserve order
TC-19	Extreme margins	10000, 9999, -10000	Stable finite normalized output	Passed; first class remains largest
TC-20	Invalid temperature	Zero, negative, boolean, NaN	Refuse calibration parameter	All invalid values refused
TC-21	Study partial output	Failed/missing candidate rows	No paired score substitution	Coverage refusal passed
TC-22	Original CryptoBERT probes	Five authored crypto sentences	Outputs vary; retain raw results	Both directions; neutral probe failed
TC-23	Delivery admission	Failed, stale or wrong-commit receipt	Do not deliver	Nine guard cases passed
TC-24	Checkout content drift	Changed bytes beyond newline translation	Refuse restoration	Content and path refusal passed

TC-15 and TC-19–20 are controlled fixtures; TC-22 is a model diagnostic. Suite receipts remain separate.

Appendix D

Repository and Reproduction

D.1 Source and Artifacts

Repository: https://github.com/Ahmed-Javaid/Investify-FYP-2027. The current delivery configuration selects the reviewed report/presentation. Dated receipts certify their exact code/input/output snapshots. Original source and experimental freezes remain preserved; subsequent revisions receive new identities.

D.2 Demo Setup

Use Git and Python 3.11. Install `code/requirements-demo-runtime.txt`, restore reviewed checkout bytes and run the Streamlit entry described in Chapter 7. The complete team guide is `docs/team/MID1_TEAM_GUIDE.md`; technical references and the agent handoff are separate under `docs/technical` and `docs/handoffs`. Source code comes from GitHub; the compact groupmate ZIP contains report, slides, rubric and the consolidated guide.

D.3 Scientific Reproduction

The original fresh snapshot is `code/artifacts/scientific_runs/alpaca_revision_conservative_20261012_run01`. The new calibration correction is `code/artifacts/scientific_runs/temperature_calibration_20261013_run01`. The temperature builder replays immutable model/scaler/margin inputs and creates new records; it does not replace the original calibration. The social study retains original models, plan, reference labels and paired output receipts. The added sanity probes have a separate receipt.

D.4 Build and Delivery

The institutional class/logo remain unchanged. Compile sources into a new scratch directory, render and inspect the resulting pages, then publish a new source/PDF receipt. CI packages only matching reviewed documents and clean same-commit prerequisite receipts. Installation and extraction checks establish their scoped engineering behavior; production serving and participant acceptance remain separate milestones.

Appendix E

Design Model Reference

The use-case, architecture, class, file-schema, sequence, activity, state, risk, deployment and timeline figures are presented beside their explanations in Chapters 4–7 and 10. They are not repeated with unlabeled connectors in this appendix. Native editable TikZ sources are retained under `report/figures`; code-linked responsibilities appear in the requirements matrix.

Appendix F

Meeting Evidence

Meeting records are not included in this version.

Appendix G

Document Control and Responsible Development

G.1 Document Control

Table G.1: Report control information.

Field	Value
Project	Investify, F26-003-Investify
Assessment	FYP-1 Mid-Evaluation, Fall 2026
Report date	11 October 2026; revision/testing retain actual 13–14 October dates
Version	1.0, Mid-1 technical and presentation revision
Authors	Ahmed Javaid, Rafay Ul Haq, Soban Ahmad
Supervisor	Mr. Irfan Ullah
Delivery	Source, PDF review, test and archive identities retained separately

G.2 Development and Reference-Construction Disclosure

Development used coding assistants for implementation suggestions, test construction, diagrams and prose revision. The team integrated the delivered modules and accepts responsibility for understanding, attribution and defence. External data/models/software are cited and original-source identities are retained.

The social reference labels were constructed in a single model-assisted annotation pass using predefined target-equity sentiment and exclusion rules. Prompt/input/raw-output provenance is retained. No independent human annotation or inter-rater reliability was measured; reported social scores are agreement with this reference. The teacher construction pilot uses actual generated prose with deterministic factual JSON assembled from the record. Available configuration-level identity is retained; the exact serving revision is unavailable. Student inference and adaptation remain later work.

G.3 Dependencies and Scope

Table G.2: External components and project responsibilities.

Component	Use	Project responsibility
FinBERT / source corpora	News sentiment and historical inputs	Timestamp/coverage audit, versioning and lawful use
XGBoost / native SHAP	Model and attribution	Causal fitting, calibration diagnosis and numerical replay
CryptoBERT / Cardiff	Social candidate study	Original bytes, mappings, frozen split and scoped comparison
Grable–Lytton	Published tolerance scoring	Capacity reconciliation, wording disclosure and local validation
Streamlit / Python	Viewer and test/runtime tooling	Current-state behavior, setup and failure handling
GitHub Actions	Automated validation/delivery	Same-source gates, retention and verified artifacts

G.4 Submission and Evaluation Coverage

The report contains vision/research, numbered requirements, detailed use cases, project-specific architecture/data/behavior models, an implementation/manual, individual test cases, metrics, member delivery and a dated plan. Signature spaces are provided for printing. The meeting-evidence status appears in Appendix F. The current software uses file/session contracts; no relational database or brokerage execution is claimed.

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